



OPEN Optimized driver fatigue detection method using multimodal neural networks

Shengli Cao¹, Peihua Feng²✉, Wei Kang³, Zeyi Chen² & Bo Wang¹

Driver fatigue is a significant factor contributing to road accidents, highlighting the need for precise and reliable detection systems. This study introduces a comprehensive approach using multimodal neural networks, leveraging the DROZY dataset, which includes physiological and facial data collected under sleep deprivation conditions. Two advanced neural network models were developed and evaluated: a multimodal feature combination model and a multimodal feature coupled model. The multimodal feature coupled model, which is the highlight of this study, uniquely integrates various data types, such as electroencephalograms (EEG), electrocardiograms (ECG), and facial images, through a sophisticated coupling mechanism. In this model, the features from different modalities are not merely combined; instead, they serve as mutual weights, influencing each other's contribution to the final prediction. This dynamic interaction between features enhances the model's ability to capture complex patterns associated with driver fatigue, leading to superior performance. The model achieved an outstanding accuracy of 98.41%, precision of 98.38%, recall of 98.39%, and an F1-score of 98.38%, demonstrating its exceptional capability in accurately detecting driver fatigue. In comparison, the multimodal feature combination model also performed well, with an accuracy of 94.87%, precision of 95.18%, recall of 95.04%, and an F1-score of 95.00%. To ensure robust decision-making in real-world applications, a majority voting strategy was employed in the decision phase, aggregating predictions from multiple classifiers to provide consistent and reliable fatigue warnings. These results highlight the advantages of the multimodal feature-coupled model in addressing the challenges of driver fatigue detection, making it a valuable tool for enhancing road safety through advanced, efficient monitoring systems in vehicles.

Keywords Driver fatigue, Feature combination, Multimodal feature coupled mode, Road safety

Driver fatigue, characterized by diminished alertness and prolonged reaction times following extended periods of intensive driving, poses a significant hazard that can lead to accidents, thereby jeopardizing the safety of both the driver and the public^{1,2}. Consequently, identifying and predicting fatigue are imperative for enhancing road safety. Approaches to fatigue detection are generally divided into two main categories: subjective and objective methodologies³.

To begin with, subjective techniques involve psychological evaluations through questionnaires or interviews, exemplified by instruments such as the Karolinska Sleepiness Scale (KSS)⁴, the Stanford Sleepiness Scale (SSS)⁵, and the Manchester Driver Behaviour Questionnaire⁵. Although these methods are easy to administer, they are nonetheless susceptible to individual differences and interpretation biases, which may compromise their reliability. As a result, subjective approaches are often employed in a supplementary capacity within research contexts to validate the accuracy of alternative fatigue detection strategies.

In contrast, objective methodologies utilize external devices to collect data during driving, thereby enabling algorithmic assessments of fatigue status. Due to their superior precision and independence from driver subjectivity, these techniques have emerged as the focal point of research in fatigue detection. These objective approaches analyze various data types while driving, including metrics related to vehicle behavior, physiological signals, facial expressions, and vocal patterns, each offering valuable insights into the driver's level of fatigue⁶.

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Vehicle behavior

For instance, indicators of vehicle behavior, such as velocity, acceleration, yaw angle, and lane deviation, serve as critical markers of driver fatigue^{7,8}. In this context, researchers categorize states of driver fatigue based on these variables. For example, Riera et al.⁹ monitor the centroid of convex hulls within the forward lane area from driving footage to identify lane deviations. Similarly, Chen et al.¹⁰ utilize the power spectral density of lane deviations and levels of driver abnormality to issue warnings of hazardous driving. Moreover, McDonald et al.¹¹ employ dynamic Bayesian network models, incorporating variables such as steering angle, pedal position, velocity, and acceleration to detect signs of fatigue. Furthermore, Li et al.¹² analyze the approximate entropy of steering wheel angle sequences, applying Long Short-Term Memory (LSTM) units to extract features indicative of fatigue.

Physiological signal

In addition to vehicle behavior, physiological signals are crucial for driver fatigue detection, as they provide direct insights into the driver's cognitive and physical state. Key signals include Electroencephalogram (EEG), Electrocardiogram (ECG), and Electromyogram (EMG), each offering unique perspectives on fatigue. To illustrate, EEG signals, which capture the brain's electrical activity, are widely used to monitor cognitive states and detect fatigue. Specifically, EEG features, particularly in the frequency domain, have been central to these efforts¹³. For instance, Lu et al.¹⁴ developed a drowsiness recognition model using deep residual networks, focusing on enhancing EEG feature fusion while eliminating artifacts, thereby improving classification precision. Likewise, Guarda et al.¹⁵ introduced a Capsule Neural Network (CapsNet) that leverages EEG channel imagery to better capture spatial features, significantly improving detection accuracy over traditional CNNs. Additionally, Houshmand¹⁶ utilized continuous wavelet transforms and Morlet functions to extract key EEG features, emphasizing frequency domain characteristics critical for fatigue detection.

On the other hand, ECG signals are vital for monitoring heart rate variability and other cardiac features associated with fatigue. These features provide insights into the driver's physiological state under varying conditions. For example, Murugan et al.¹⁷ focused on ECG features like heart rate variability, applying ANOVA and PCA for feature selection, followed by SVM and KNN classifiers to enhance detection accuracy. Furthermore, Gromer et al.¹⁸ highlighted the importance of heart rate variability metrics in detecting fatigue. Similarly, Shahrudin et al.¹⁹ explored ECG features related to drowsiness, demonstrating their relevance in identifying subtle physiological changes indicative of reduced alertness.

Moreover, EMG signals, which monitor muscle activity, are essential for detecting physical fatigue during prolonged driving. For instance, EMG features, such as muscle activation patterns, have been effectively used to identify fatigue. In this regard, Fan et al.²⁰ integrated EMG features with a Gated Recurrent Unit (GRU) classifier, achieving high accuracy in detecting fatigue-related muscular strain. Additionally, Yaacob et al.²¹ combined EMG features with EEG and ECG signals to significantly enhance overall detection accuracy in multi-modal settings. Furthermore, Shahrudin et al.¹⁹ demonstrated the utility of EMG in monitoring physical fatigue-related driving behaviors.

Facial features

In addition, detection methods based on facial features are crucial for driver fatigue detection, employing machine vision technologies to analyze attributes indicative of drowsiness. Specifically, these methods track metrics like gaze direction, eye aspect ratio (EAR), blink frequency, percentage of eyelid closure over time (PERCLOS), mouth aspect ratio (MAR), head movement, and head rotation angle, providing observable indicators of the driver's alertness.

Notably, the PERCLOS standard, established by the Carnegie Mellon Institute²², is a foundational metric for objectively measuring fatigue through eye closure analysis. For example, Zhuang et al.²³ used segmented networks to assess eye openness and calculate PERCLOS values, demonstrating the effectiveness of eye-based metrics in fatigue detection. Furthermore, facial expression analysis also plays a significant role. For instance, Adhinata et al.²⁴ combined facial feature extraction with machine learning classifiers, achieving 94.68% accuracy.

Another key marker for fatigue is head movement. In this context, Ansari et al.²⁵ tracked head posture changes using the XSENS motion capture system and applied a deep learning network for fatigue assessment, highlighting the importance of head movement as a fatigue indicator. Moreover, recent advancements include deep learning techniques, such as the use of Convolutional Neural Networks (CNNs) by Sharma et al.²⁶ to analyze yawning frequency, contributing to cost-effective solutions in fatigue detection.

However, environmental factors, such as lighting and camera angles, can affect the accuracy of facial analysis. To address these challenges, Sinha et al.²⁷ explored deep learning for facial expression analysis under varying conditions. Similarly, Zhao et al.²⁸ achieved a 93.623% accuracy rate using an EM-CNN model based on facial characteristics. In addition, You et al.²⁹ introduced a real-time algorithm with a 94.32% accuracy rate by analyzing facial motion entropy, showcasing the potential of advanced facial feature extraction methods in fatigue detection.

Voice features

Furthermore, voice feature-based fatigue detection presents a non-intrusive approach by analyzing speech signal parameters such as speech speed, pitch, volume, and intonation to evaluate fatigue. In particular, Shiomi et al.³⁰ investigated the relationship between speech signals and mental fatigue through nonlinear dynamics, identifying correlations with chaotic attractor trajectories. Building upon this, Iwahori et al.³¹ created a novel framework for detecting fatigue from speech by integrating nonlinear dynamics analysis and machine learning, effectively classifying fatigue states with feature vectors derived from speech dynamics.

Multiple physiological and behavioral features

While the aforementioned studies have provided significant insights into driver fatigue detection, much of the research has focused on single data types, which may lead to potential inaccuracies due to the complexity of human physiology and individual differences. To address these limitations, recent studies have increasingly adopted integrative approaches that combine multiple physiological and behavioral measures, aiming for more accurate and reliable assessments of fatigue states.

For instance, physiological signals like EEG, ECG, and EMG have been extensively used to assess driver fatigue, as they are crucial for understanding the driver's mental state^{32,33}. Building on this, Fang et al.³⁴ advanced this approach by converting EEG data into multispectral images and using deep recurrent convolutional neural networks (R-CNNs) to analyze spatial, spectral, and temporal structures, achieving high classification accuracy. Moreover, Zhao et al.³⁵ developed a learning system that merged ECG signals with video data, significantly enhancing the accuracy of alertness state differentiation by integrating both physiological and behavioral data.

Additionally, behavioral signal-based approaches, particularly those focusing on facial features, have also been extensively explored³⁶. Specifically, machine vision techniques capture changes in facial attributes such as gaze direction, eye aspect ratio (EAR), and percentage of eyelid closure over time (PERCLOS). For example, Zhuang et al.²³ used segmented networks to assess eye openness and calculated PERCLOS values to determine fatigue levels. Similarly, Adhinata et al.²⁴ utilized FaceNet for facial feature extraction and combined it with K-Nearest Neighbors (K-NN) and Support Vector Machine (SVM) classifiers, achieving 94.68% accuracy.

Recognizing the limitations of single data types, recent research has focused on multimodal approaches that combine various physiological and behavioral signals³⁷. For instance, Perkins et al.³⁸ developed hybrid systems combining physiological signals (EEG, EMG, EOG, ECG) with video data, leading to effective drowsiness detection. Similarly, Raza et al.³⁹ employed a Logistic Regression Random Forest Classifier (LR-RFC) using smartphone motion sensor data, achieving a 99% accuracy rate. Furthermore, Du et al.⁴⁰ introduced a multimodal fusion recurrent neural network (RNN), combining heart rate and facial features from RGB-D cameras, further demonstrating the effectiveness of integrating physiological and visual data for fatigue detection.

Feature engineering plays a crucial role in determining the performance of neural networks. Beyond selecting diverse types of features, numerous scholars have introduced advanced mathematical techniques to improve feature extraction and, consequently, enhance neural network performance. For example, Wang et al.⁴¹ proposed a self-adaptive gradient projection algorithm that enhances the efficiency of feature extraction by managing high-dimensional data in sparse signal reconstruction. Similarly, Fu et al.⁴² introduced singular integral techniques to improve data transformation, allowing for a better capture of the complex relationships between parameters. Zhang et al.⁴³ contributed a structure-preserving algorithm for singular value decomposition, ensuring data integrity during feature extraction, while Li et al.⁴⁴ advanced the classification of dynamic systems, providing more refined methods to handle complex multimodal data. These methodologies collectively improve the way features are extracted, fused, and analyzed, leading to more reliable and accurate driver fatigue detection models.

Despite these advancements, many multimodal fatigue detection models still rely on extracting features from different modalities and then simply concatenating them for classification. While multifeature fusion methods do improve detection accuracy by integrating various feature advantages, they still face substantial challenges due to their inherent computational complexity. These approaches require the collection and analysis of multiple parameters and the development of intricate algorithms to combine these parameters into a cohesive fatigue assessment index. Moreover, the complex interrelations among different parameters and modalities demand extensive experimental research and data analysis to formulate effective models. Traditional multimodal models generally use separate feature extractors for distinct feature modalities, which are then combined before classification to detect driver fatigue. However, this methodology often falls short in addressing the intricate interactions between different modal features.

To overcome these limitations, this study introduces a novel method for fatigue detection that interlinks different modal features in a more integrated manner. Specifically, this research focuses on facial data and physiological signals related to fatigued driving. An innovative approach is employed where an image encoder extracts facial data features, and a time series encoder processes physiological signal features. In this model, the features from different modalities are not merely combined; instead, they serve as mutual weights, influencing each other's contribution to the final prediction. This dynamic interaction between features enhances the model's ability to capture complex patterns associated with driver fatigue, leading to superior performance. The features are multiplied together in a coupling process that allows them to mutually influence each other, effectively capturing the complex interdependencies between the different types of data. Finally, the resulting coupled features are then input into a classifier, establishing a new multimodal feature-coupled neural network. This network is uniquely designed to predict driver fatigue, meeting the complex requirements of feature engineering in fatigue detection deep learning algorithms.

Dataset and pre-processing

Dataset

The "ULg Multimodality Drowsiness Database" (DROZY), developed by the Laboratory for Signal and Image Exploitation (INTELSIG) at the University of Liège (ULg), Belgium, is a comprehensive and diverse dataset designed to support research in the field of drowsiness monitoring⁴⁵. This multimodal dataset is essential for the development and evaluation of detection systems and algorithms aimed at addressing driver fatigue and drowsiness.

DROZY includes data collected from 14 young, healthy participants (3 males, 11 females), each of whom underwent three consecutive 10-minute psychomotor vigilance tests (PVTs). These tests were conducted under conditions of increasing sleep deprivation, induced by acute and prolonged wakefulness. The database provides

meticulously time-synchronized data for each subject and each PVT session, encompassing the following components:

1. *Karolinska Sleepiness Scale (KSS) Scores* These subjective, self-reported drowsiness levels range from 1 (Extremely alert) to 9 (Very sleepy, great effort to keep awake, fighting sleep), and are recorded at the beginning of each PVT session.
2. *PVT data* This includes stimulus times and corresponding reaction times, measured in milliseconds, providing objective performance data related to alertness.
3. *Polysomnography (PSG) signals* The PSG dataset includes signals from 5 Electroencephalogram (EEG) channels (Fz, Cz, Pz, C3, and C4, all referenced to A1), 2 Electrooculogram (EOG) channels (horizontal and vertical), Electrocardiogram (ECG), and Electromyogram (EMG). These signals are sampled at 512 Hz and are stored in .edf format, offering high-resolution physiological data.
4. *Kinect v2 sensor videos* Each PVT session is accompanied by near-infrared (NIR) intensity videos captured by the Kinect v2 sensor. These videos have a resolution of 512×424 pixels, with grayscale values of 8 bits, and are stored in .mp4 format.

Additionally, the database includes both manually and automatically obtained annotations of facial landmarks, providing 2D and 3D annotations for a comprehensive set of frames.

In this study, the PSG signals, Kinect v2 sensor videos, and KSS scores from the DROZY database were primarily utilized to train neural network models for predicting driver fatigue. The integration of these specific data types-physiological signals from PSG, visual information from Kinect videos, and subjective drowsiness assessments from KSS scores-enables a comprehensive approach to modeling and understanding the multifaceted nature of driver fatigue. The utilization of such diverse and synchronized data sources offers a unique opportunity to explore the complex interplay between physiological, behavioral, and subjective factors in the onset and progression of driver fatigue.

This holistic approach to data integration allows for a more robust and accurate prediction model, as it captures the various dimensions of fatigue that contribute to impaired driving performance. By leveraging the DROZY database, researchers can advance the development of sophisticated detection systems that could potentially enhance road safety by preventing fatigue-related accidents.

Facial and physiological signals

Facial information plays a crucial role in predicting fatigue, as it captures a range of visual cues that are closely associated with drowsiness and decreased alertness. Key facial indicators, such as eye closure, blink rate, head position, and facial expressions, provide direct and observable signs of fatigue. For example, frequent blinking, prolonged eye closure, and yawning are well-established markers of tiredness that can be detected through facial analysis. By integrating facial information into fatigue prediction models, it is possible to monitor these subtle yet significant changes in real-time, allowing for a more accurate and timely assessment of a driver's fatigue level. The visual nature of facial data also complements physiological signals, offering a holistic approach to detecting fatigue that combines both external behaviors and internal states. This makes facial information a valuable component in developing robust systems for predicting and mitigating fatigue-related risks.

EEG, ECG, EMG, and EOG signals are critical physiological indicators that provide valuable insights into a person's state of alertness and fatigue, making them essential for detecting driver fatigue effectively. Understanding these signals is crucial, as they each reflect different aspects of the body's response to fatigue. EEG measures the electrical activity of the brain, capturing spontaneous brain activity through electrodes placed on the scalp. In a normal, alert state, EEG signals typically exhibit a mix of alpha (8–12 Hz) and beta (12–30 Hz) waves, which indicate a wakeful and active brain. However, during fatigue, particularly as drowsiness sets in, there is an increase in theta waves (4–7 Hz) and a decrease in alpha activity, signaling reduced alertness as the brain shifts towards a more sleep-like state.

ECG records the electrical activity of the heart, playing a crucial role in monitoring heart rhythms and detecting the physiological effects of fatigue. Normally, the ECG displays a regular pattern of P, QRS, and T waves, corresponding to the heart's rhythmic contractions, with a typical heart rate of 60–100 beats per minute (bpm). When fatigue sets in, heart rate variability may increase, and there may be a slight reduction in heart rate. Additionally, subtle changes in the amplitude or shape of the QRS complex and T wave may occur, reflecting the body's reduced physical and mental state.

EMG captures the electrical activity produced by skeletal muscles, providing information on muscle health and function. In a normal state, EMG signals show regular bursts of activity corresponding to voluntary muscle contractions, with stable amplitude and frequency. However, as muscles become fatigued, the EMG signals typically show a decrease in amplitude and frequency, indicating the muscles' reduced capability to sustain contractions. This reduction in signal strength reflects the onset of muscle fatigue.

EOG measures the electrical potential generated by eye movements, capturing the difference in potential between the front and back of the eye. This signal is particularly useful for monitoring eye movements, including blinks and gaze direction, which are key indicators of drowsiness. In an alert state, EOG signals show distinct, rapid eye movements with clear, sharp deflections corresponding to regular blinks and saccades (quick eye movements). As fatigue increases, these signals tend to slow down, with longer, more drawn-out blinks and less frequent eye movements, reflecting decreased alertness.

To illustrate these differences, signals were plotted from a subset of participants in both normal (alert) and fatigued (drowsy) states. As shown in Fig. 1, the comparison of these physiological signals revealed significant differences between the two states. For instance, EEG signals in the alert state exhibit higher variability and amplitude, indicating more active brain function, whereas the drowsy state is characterized by reduced amplitude

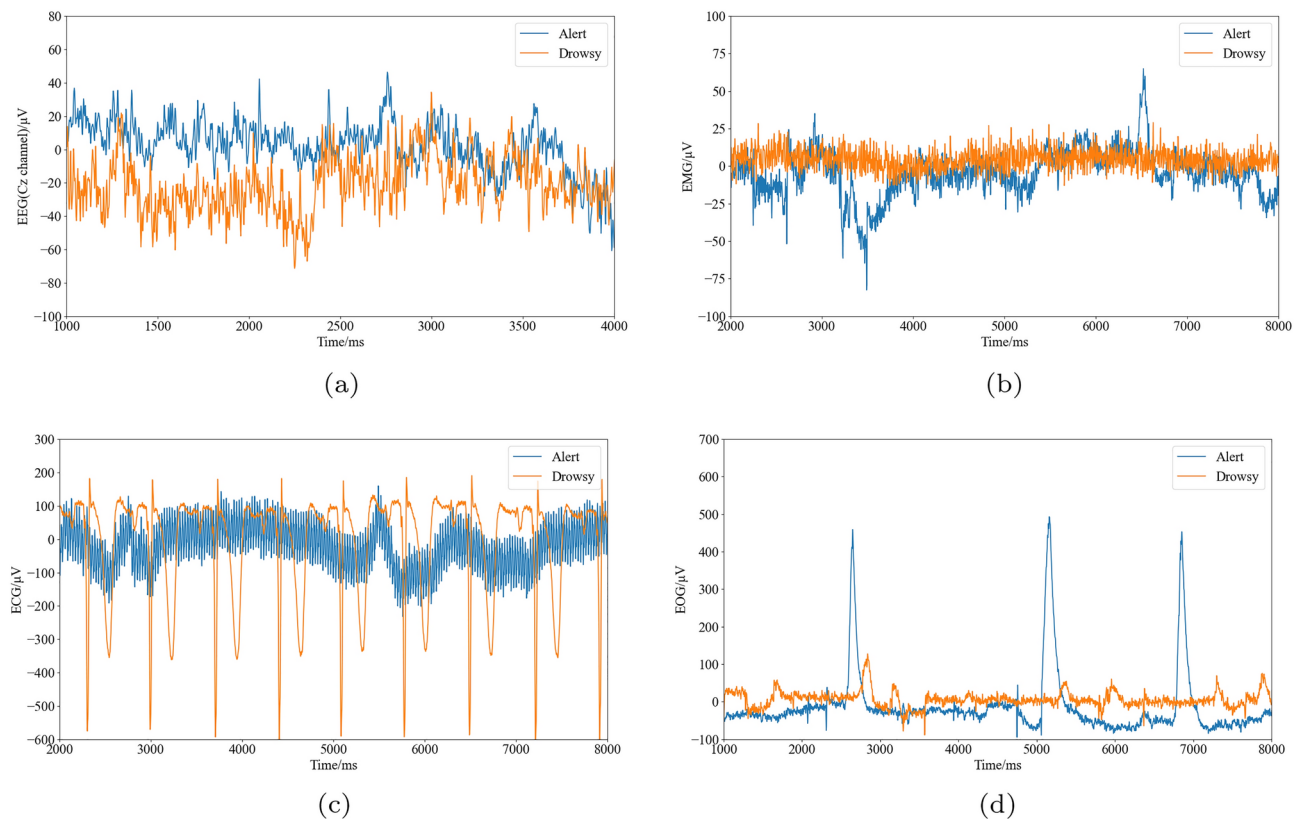


Fig. 1. Comparison of physiological signals in alert and drowsy states.

and more uniform waveforms, reflecting decreased cognitive activity. Similarly, EMG signals demonstrate higher muscle activity in the alert state, with more pronounced peaks, compared to the lower amplitude and more uniform signals observed in the fatigued state, which suggest muscle relaxation. EOG signals in the alert state are marked by sharper and more frequent eye movement peaks, contrasting with the slower and less distinct peaks seen in the drowsy state. Lastly, ECG signals reveal a more regular and robust heartbeat in the alert state, while the drowsy state shows irregularities and lower overall amplitude, indicating changes in heart rate and rhythm that often accompany fatigue.

These physiological signals collectively provide a comprehensive picture of the body's response to fatigue. By analyzing the changes in EEG, ECG, EMG, and EOG signals, it becomes possible to detect early signs of driver fatigue, enabling timely interventions to prevent accidents and enhance road safety. The clear differences observed between normal and fatigued states across these signals further validate their use in fatigue detection systems, making them reliable indicators for assessing driver fatigue.

Dataset pre-processing

The preprocessing of the dataset for this study entailed multiple steps to ready the data for training neural networks. The dataset comprises recordings from 14 subjects who participated in three psychomotor vigilance tests (PVTs), each documented through near-infrared (NIR) intensity videos at 30 frames per second (fps) and a resolution of 512×424 pixels, focusing primarily on the subjects' upper body and facial information.

As shown in Fig. 2, to emphasize facial information critical for driver fatigue assessment, the study leveraged OpenCV for frame extraction from the 30 fps videos and the dlib library for facial region cropping. Each facial region was resized to a consistent dimension of 75×75 pixels, acknowledging that the videos, being grayscale, necessitate only one channel, thus resizing the images to (1 channel, 75×75 pixels). Due to missing videos in the dataset, a total of 36 effective videos were used, yielding approximately 496,529 facial images for neural network training, constituting the facial expression modality data.

In addition to the PVTs, Polysomnography (PSG) signals were recorded, including five EEG channels, two EOG channels (horizontal and vertical), and one channel each for ECG and EMG. These PSG signals were synchronized with the Kinect v2 sensor videos. The alignment between Polysomnography (PSG) signals and facial images was achieved by using each frame of the facial images as the reference timestamp. For each frame captured during the experiment, a corresponding 1-s time window of PSG signals was extracted, with the end of this window aligning with the timestamp of the facial image. This means that for every facial image, the PSG data used as the corresponding sample represents the physiological activity occurring in the 1 s leading up to the captured image. This process resulted in localized PSG signals comprising 512 timesteps across nine channels, ensuring that the PSG signals were closely aligned with the precise moment the facial expressions were captured.

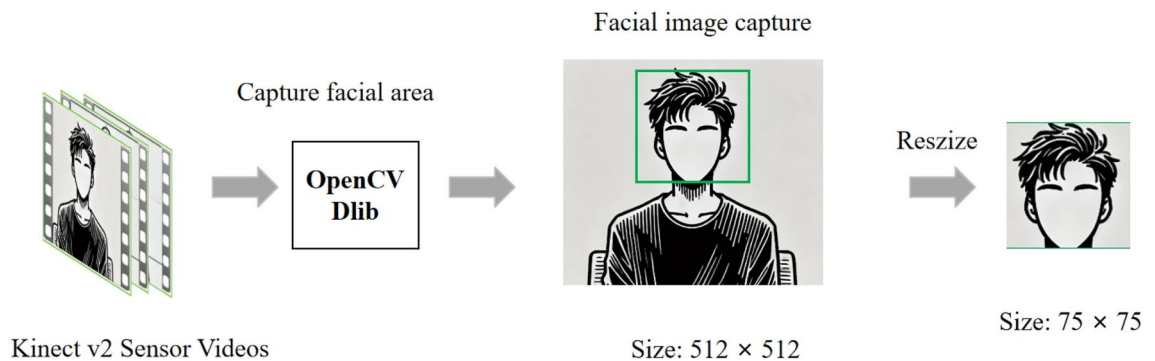


Fig. 2. Workflow for facial image extraction, cropping, and resizing from kinect v2 sensor videos for driver fatigue analysis.

This allows for a coherent analysis of the relationship between physiological responses and facial cues in the context of driver fatigue detection.

To process the facial images, we stack all the frames within each time window and apply Averaging Pooling Across Frames to merge these images into a single sample. By averaging the pixel values across the stacked frames, we create a single representative image for each time window. This method has several key advantages: First, it is simple to implement, requiring only the stacking and averaging of frames without the need for complex architectures or additional processing steps. Second, it captures more temporal information, as combining multiple frames allows the model to retain subtle changes in facial expressions over time, leading to a more comprehensive representation than relying on a single frame. Lastly, the method has a low computational cost, as averaging across frames is computationally efficient compared to more complex methods like temporal sequence modeling, making the data processing faster while preserving the key temporal context.

At the same time, subjects self-reported their sleepiness levels using the Karolinska Sleepiness Scale (KSS), which ranges from 1 (extremely alert) to 9 (very sleepy, struggling to stay awake). The KSS scores were then classified into three categories for neural network training: Alert (scores 1–3), Low Vigilance (scores 4–6), and Drowsy (scores 7–9). The fatigue level for each sample was determined based on the KSS scores at the precise moment of the corresponding facial image. By aligning the PSG signals and KSS scores with the facial images, we ensured that the physiological data, facial expressions, and subjective fatigue ratings were synchronized.

By integrating facial expression data, localized PSG signals, and categorized KSS scores, this study aims to construct a comprehensive model for accurately detecting driver fatigue levels. After completing the preprocessing and alignment steps, a total of 496,529 samples (35,446 samples per person) were generated for use in training and evaluating the neural network models. This combined approach enables a robust and efficient method for processing data in driver fatigue detection while minimizing computational complexity.

Dataset splitting

In this study, to ensure the reliability of the model evaluation process, we adopted a subject-based data splitting method, rather than using image samples. This splitting approach ensures the independence and uniqueness of the training, validation, and test sets, preventing evaluation bias caused by sample repetition. Specifically, each second of data contains 30 frames of images, with each frame corresponding to a data sample. If image-level splitting were directly applied, it could result in adjacent frames appearing in both the training and validation sets simultaneously, thereby affecting the generalization capability of the model. Therefore, an image-based splitting approach is not suitable for the machine learning model used in this study. To address this issue, we opted for a subject-based splitting method, where subjects are randomly selected for training and evaluation in each training, validation, and testing process.

In the specific process of model training, validation, and testing, the dataset comes from 14 subjects. During each training session, the sequence of the 14 subjects is first randomly shuffled and labeled as $s_1, s_2, \dots, s_{14}$. This is done to ensure that the dataset used in each training and evaluation process is independent and diverse, avoiding biases that may arise from relying on specific subjects' data. Next, 13 of these 14 subjects are randomly selected for training and validation set splitting. Specifically, the data from these 13 subjects form the dataset $D_{\text{train+val}}$, which is then split into training and validation sets in a 9:4 ratio. The training set consists of data from 9 subjects, used for model learning and parameter optimization; the validation set includes the data from the remaining 4 subjects, used for tuning (tuning) the model's hyperparameters, ensuring the model's generalization to unseen data.

The remaining one subject serves as an independent test set and is completely excluded from the training and validation process. The independent test set design can truly reflect the model's performance on unseen data, and the results from the test set indicate the model's actual generalization ability. During each independent training, validation, and testing process, the test set's data remains fully independent to ensure that the model does not overfit the data from specific subjects. The benefit of this design is that it allows the model to demonstrate strong adaptability under different data distributions. The training set contains 318,996 samples, the validation set contains 142,087 samples, and the test set contains 35,446 samples.

Mathematically, let the dataset D be composed of data from 14 subjects, with each subject's dataset represented as $D^{(s)}$, where s is the subject number. After random shuffling and labeling, the dataset is split as follows: The data from 13 randomly selected subjects $D_{\text{train+val}}$ are used for training and validation set splitting, where the training set contains data from 9 subjects:

$$D_{\text{train}} = D^{(s_1)} \cup D^{(s_2)} \cup \dots \cup D^{(s_9)}. \quad (1)$$

The validation set consists of the remaining 4 subjects' data:

$$D_{\text{val}} = D^{(s_{10})} \cup D^{(s_{11})} \cup D^{(s_{12})} \cup D^{(s_{13})}. \quad (2)$$

Finally, the remaining one subject is used as the independent test set:

$$D_{\text{test}} = D^{(s_{14})}. \quad (3)$$

During each training session, the model is first trained on the training set D_{train} , which covers a large number of samples from 9 subjects and helps the model learn effective features from diverse samples. Then, tuning is performed on the validation set D_{val} , where the model's hyperparameters are adjusted based on its performance on the validation set to better generalize to unseen data and avoid overfitting to the specific training data. Finally, the model's performance is evaluated on the test set D_{test} , which remains completely independent during the training and validation stages, and the test results truly reflect the model's generalization ability.

To further enhance the robustness of the model evaluation, the study conducted 10 independent training, validation, and testing processes. In each independent training session, a different set of 13 subjects was randomly selected for training and validation, while the remaining subject was used as the independent test set. This method, by randomly selecting and splitting the subject data multiple times, ensures the diversity and independence of the model evaluation results, preventing the model from overfitting to a specific data combination. After 10 independent training and testing sessions, the final model performance is calculated as the average of the results from each test. Let $\text{metric}^{(j)}$ represent the performance metric for the j -th independent test, where $j = 1, 2, \dots, 10$ corresponds to each of the 10 independent training and testing sessions. The overall performance metric Performance metric is calculated as:

$$\text{Performance metric} = \frac{1}{10} \sum_{j=1}^{10} \text{metric}^{(j)}. \quad (4)$$

In this way, the overall performance metric is based on the average of the results from the 10 independent tests. This design avoids accidental bias in a single test, ensuring the fairness and reliability of the model evaluation. Each independent training, validation, and testing process is fully independent and provides a realistic reflection of the model's robustness under different data combinations. Ultimately, through the results of the 10 independent tests, the study can comprehensively assess the model's robustness and generalization ability, ensuring its ability to handle different data distributions and variations in practical applications. This method of data splitting and multiple independent tests effectively prevents the model from relying on a specific data distribution and improves the model's adaptability and performance when facing new data.

Proposed methodology

This section outlines the proposed methodology for detecting driver fatigue using a multimodal approach that integrates facial data and physiological signals. The workflow of the methodology, as depicted in Fig. 3, provides a step-by-step overview of the entire process—from data preprocessing to feature extraction, model development, evaluation and model application. The methodology begins with preprocessing the input data to ensure it is suitable for further analysis. Subsequently, relevant features are extracted using specialized encoders tailored for both image and time-series data. These extracted features are then processed through two distinct models: a feature combination model and a feature coupling model. The latter introduces a novel approach that allows features from different modalities to interact dynamically, thereby enhancing the model's ability to capture complex patterns associated with driver fatigue.

Then, the models are trained and evaluated using a robust cross-validation technique and further optimized through hyperparameter tuning to ensure the highest level of accuracy and reliability.

In the model application and decision-making phase, the majority voting strategy is employed to combine the predictions from multiple classifiers. This strategy plays a critical role in ensuring that the system provides reliable and consistent alerts based on the detection of fatigue. Specifically, the system analyzes the data over consecutive time steps and applies majority voting to the predictions of multiple classifiers. If more than half of the classifiers output a “positive” state, indicating the detection of fatigue, a warning is issued to the driver.

This robust methodology ensures that the system not only captures real-time driver fatigue but also achieves high accuracy and reliability through advanced feature integration, model tuning, and the application of the majority voting strategy in the decision-making phase.

To achieve this, the LSTM and ResNet18 models were selected due to their specific strengths in addressing the requirements of driver fatigue detection, which involves both image data (facial features) and time-series data (physiological signals). ResNet18 was chosen as the backbone for image feature extraction because of its proven effectiveness in handling image data, particularly in tasks involving deep convolutional neural networks

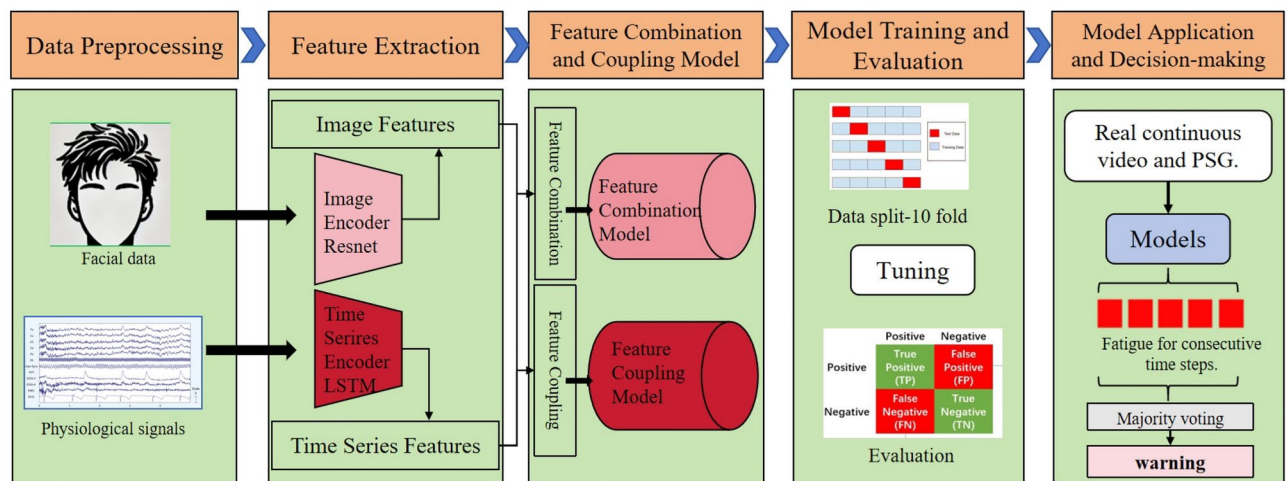


Fig. 3. Workflow diagram of the proposed methodology.

(CNNs). As part of the Residual Networks (ResNet) family, ResNet18 addresses the vanishing gradient problem commonly encountered in deep networks by introducing skip connections, which allow gradients to flow more directly through the network layers. This design enables ResNet18 to maintain high accuracy even with deeper architectures, making it particularly suitable for tasks requiring detailed feature extraction from images. For driver fatigue detection, facial features such as eye closure, head pose, and facial expressions are critical indicators. ResNet18's ability to capture these fine-grained features, while maintaining computational efficiency, makes it an ideal choice. Compared to other deeper or more complex architectures, ResNet18 offers a balance between performance and efficiency, which is crucial for real-time applications like in-vehicle monitoring systems where computational resources may be limited.

In parallel, LSTM (Long Short-Term Memory) networks were selected to process the time-series physiological data, such as EEG, ECG, and EMG signals, which are essential for detecting the internal states of the driver that contribute to fatigue. LSTMs are a type of recurrent neural network (RNN) specifically designed to handle long-range dependencies in sequential data. They are particularly effective in capturing temporal patterns and trends over time, which are crucial in understanding the evolution of fatigue in a driver. In contrast to traditional RNNs, LSTMs address the vanishing gradient problem, allowing them to retain and learn from information over longer sequences of data. This capability makes LSTMs especially well-suited for modeling the physiological signals that change gradually as a driver becomes fatigued. By leveraging LSTM networks, the model can recognize subtle temporal dynamics within the physiological data, which are indicative of driver fatigue, thereby providing a robust mechanism for early detection.

The combination of ResNet18 and LSTM models leverages the strengths of both architectures: ResNet18 efficiently extracts spatial features from facial images, while LSTMs capture the temporal dependencies in physiological signals. This multimodal approach is particularly well-suited to the complex and multifaceted nature of driver fatigue detection, where both visual cues and internal physiological states must be considered. By integrating these two models, the study aims to provide a more comprehensive and accurate detection system that outperforms models relying solely on either image or time-series data.

In summary, the selection of ResNet18 and LSTM was driven by their respective abilities to handle spatial and temporal data effectively, their robustness in addressing the specific challenges of driver fatigue detection, and their suitability for integration into a multimodal framework that combines image and physiological data for enhanced prediction accuracy. The application of the majority voting strategy ensures that the system issues alerts only when consistent and reliable evidence of fatigue is detected, further enhancing its robustness and reducing the risk of false alarms.

Basic network architecture

LSTM

In this study, long short-term memory (LSTM) networks, a variant of recurrent neural networks (RNNs), are employed for feature extraction from Polysomnography (PSG) signals. LSTMs are particularly adept at processing time-series data, such as PSG signals, which are pivotal in studies of sleep and fatigue. Designed to address the limitations of conventional RNNs—namely the vanishing gradient problem that impedes learning over long sequences—LSTMs facilitate the retention of information across extended intervals. This issue arises when gradients diminish during backpropagation through time, leading to a loss in the network's ability to learn long-range dependencies.

The essence of LSTMs lies in the cell state, which functions similarly to a conveyor belt running through the LSTM units, carrying pertinent information across the sequence. This mechanism enables the network to make informed decisions based on long-term context. LSTMs incorporate three gates—input, forget, and output—that manage the information flow within the cell state, determining the retention, update, or discard of information.

This structure allows LSTMs to effectively learn and retain information over long durations, making them particularly suitable for analyzing time-series data such as PSG signals.

Figure 4 illustrates the architecture of the Long Short-Term Memory (LSTM) network employed for processing Polysomnography (PSG) signals in this study. The LSTM network expands the input data through a series of LSTM cells, each designed to capture and learn temporal dependencies in the sequential data. Each LSTM cell processes the input data, utilizing gates such as the forget gate, input gate, and output gate to manage the flow of information and maintain long-term dependencies. The architecture integrates key components, including the cell state from the previous moment (C_{t-1}), the previous moment's output (h_{t-1}), the current input (x_t), the Sigmoid activation function (σ), and the hyperbolic tangent activation function ($\tanh$).

The LSTM calculation process is as follows:

f_t represents the output of the forget gate at the current moment:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f). \quad (5)$$

i_t represents the output of the input gate at the current moment:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i). \quad (6)$$

$\tilde{C}_t$ represents the current candidate cell state:

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C). \quad (7)$$

C_t represents the updated cell state:

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t. \quad (8)$$

o_t represents the output of the output gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o). \quad (9)$$

h_t represents the output at the current moment:

$$h_t = o_t \times \tanh(C_t). \quad (10)$$

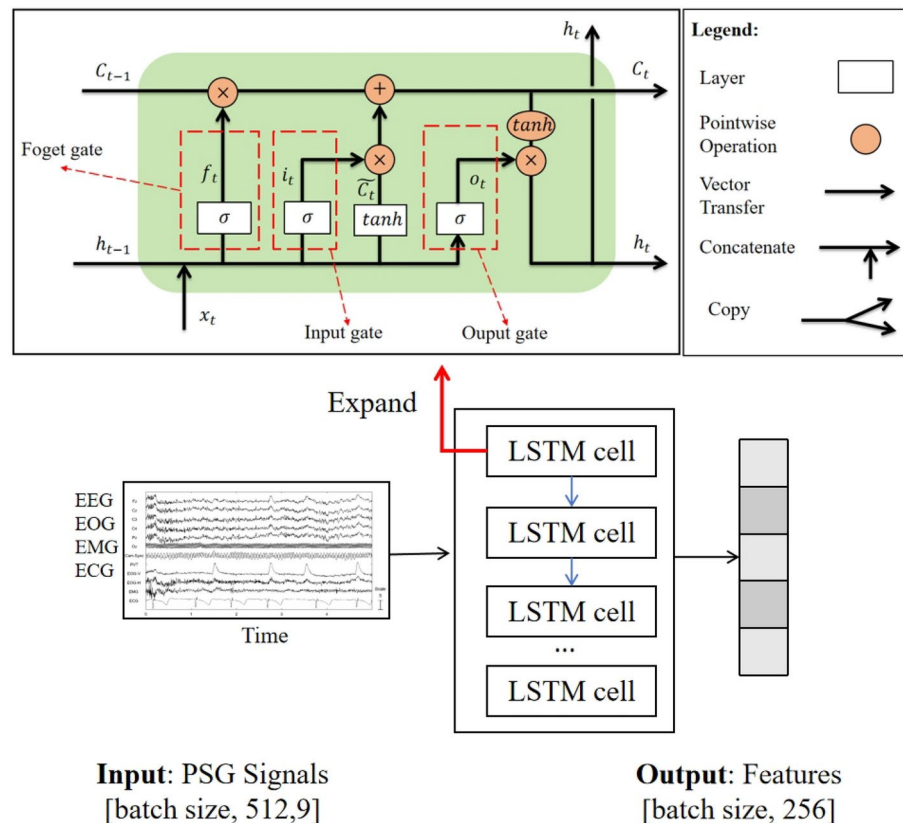


Fig. 4. Long short-term memory (LSTM) network architecture for processing PSG signals.

In these equations, W_f , W_i , W_C , W_o represent the weights of the forget gate, input gate, candidate input gate, and output gate respectively, while b_f , b_i , b_C , b_o represent the corresponding biases.

As shown in Fig. 4, the LSTM neural network takes as input a three-dimensional tensor of Polysomnography (PSG) signals, formatted as [batch size, 512 timesteps, 9 channels]. This input tensor represents a batch of physiological data sequences, where each sequence spans 512 time steps (capturing temporal dynamics), and each time step contains data from 9 different physiological channels (including EEG, EOG, EMG, and ECG signals).

As the input tensor is fed into the LSTM network, the network processes the data step by step across the time dimension. The LSTM architecture, specifically designed to handle sequential data, includes memory cells that maintain a hidden state and a cell state at each time step. These states are updated as the network processes each time step, allowing the LSTM to retain and utilize information from previous time steps, effectively capturing long-term dependencies within the sequence.

During this process, the LSTM applies a series of operations—such as input gating, forget gating, and output gating—at each time step to determine which information should be added to the cell state, which should be retained, and which should be discarded. This selective memory mechanism enables the LSTM to focus on relevant temporal patterns and ignore irrelevant noise, which is particularly important for detecting fatigue-related changes in physiological signals that may evolve gradually over time.

Once the LSTM has processed the entire sequence (512 timesteps for each batch), the network outputs a two-dimensional tensor of learned features, formatted as [batch size, 256 features]. This output tensor represents a condensed and informative set of features extracted from the original input signals. The reduction from 512 timesteps to 256 features indicates that the LSTM has successfully distilled the temporal information into a more compact representation, capturing the essential characteristics of the physiological signals relevant to fatigue detection.

These 256 features encapsulate the dynamic interactions between the different physiological channels over time, providing a rich feature set that is subsequently used in the later stages of the model to predict the driver's fatigue level. The LSTM's ability to learn and extract these features from sequential data is crucial for the model's overall accuracy, as it ensures that both short-term fluctuations and long-term trends in the physiological signals are accounted for in the fatigue prediction process.

ResNet 18

In this study, the Residual Network (ResNet) framework is utilized for image feature extraction, capitalizing on its advanced capabilities in managing deep Convolutional Neural Network (CNN) structures for computer vision tasks. ResNet effectively addresses the critical “vanishing gradient” issue prevalent in CNNs. This problem emerges when a deep network experiences a progressive decrease in gradient magnitude during backpropagation due to repeated multiplications, which impedes the network's learning capacity and results in performance saturation or decline as layers are added.

ResNet introduces an innovative approach to mitigate this issue through “skip connections,” enabling the network to incorporate multiple identity mappings (convolutional layers) while bypassing certain layers. This mechanism facilitates a more efficient training phase by simplifying the gradient flow and allowing for the scaling of the network to many layers without significantly increasing the computational burden. Initially, the network benefits from these connections by enhancing the early stages of training. Upon further training, the network's residual components are refined, allowing for a more in-depth exploration of the input image's feature space. A standard residual block, depicted in Fig. 5, comprises two to three layers interspersed with nonlinearities and batch normalization. A distinctive aspect of these blocks is the summation of a layer's input with the output of a subsequent layer, a technique now widely adopted across various neural network designs.

ResNet18, a specific ResNet model variant, is optimized for deep learning in computer vision, characterized by its 18-layer structure that includes convolutional layers, batch normalization, ReLU activation functions, and pooling layers. As illustrated in Fig. 6, ResNet18 distinguishes itself with residual blocks, or “ResBlocks,” featuring skip connections. These connections allow a layer's input to be added directly to the output of a subsequent layer. Each ResNet18 residual block generally encompasses two convolutional layers, complemented by batch normalization and ReLU activation. The skip connections facilitate the learning of an identity function, ensuring that the addition of layers does not compromise network performance. This configuration promotes direct gradient flow, simplifying the training of deeper models.

Despite its smaller scale, ResNet18 provides an efficient and computationally economical alternative to larger models like ResNet34, ResNet50, or ResNet101, without sacrificing the ability to capture complex image features.

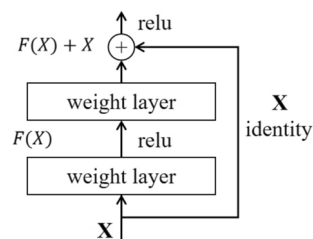


Fig. 5. Residual block.

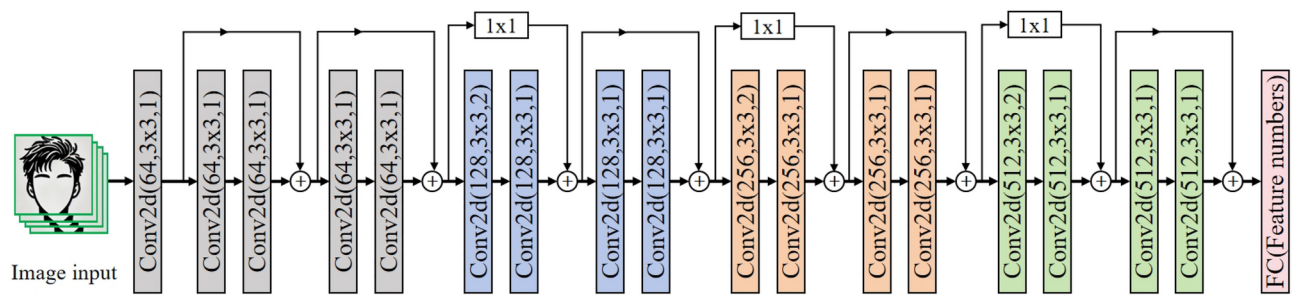


Fig. 6. Resnet architecture.

It has become a preferred choice for a range of computer vision applications, such as image classification, object detection, and facial recognition. In this research, ResNet18 served as the foundation for extracting facial features from the dataset, with modifications tailored to the specific requirements of the study.

The input to the ResNet18 network is a batch of grayscale facial images, formatted as [batch size, 1 channel, 75 pixels, 75 pixels]. Each image in this batch is a 75×75 pixel grayscale image, where the single channel represents the intensity of light for each pixel. ResNet18 processes these images through a series of convolutional layers, pooling layers, and activation functions. The initial layers of ResNet18 apply convolution operations to detect low-level features such as edges and textures. As the data progresses through the deeper layers of the network, more complex features are extracted, including facial structures, shapes, and specific patterns relevant to detecting signs of fatigue, such as eye closure or yawning.

The “residual” aspect of ResNet18 allows the network to learn more effectively by mitigating the vanishing gradient problem. This is achieved through shortcut connections that bypass one or more layers, allowing the network to maintain the flow of gradients during backpropagation. These shortcut connections enable ResNet18 to train deeper networks, thereby capturing richer and more abstract features from the input images.

As the facial images are processed through ResNet18, the network progressively reduces the spatial dimensions while increasing the depth of the feature maps, condensing the information into a more abstract form. After passing through the final layers of ResNet18, the output is a two-dimensional tensor formatted as [batch size, 256 features]. This tensor represents a set of 256 learned features for each image, encapsulating the most significant aspects of the facial data relevant to fatigue detection. These extracted features provide a condensed representation of the original facial images, capturing essential visual cues that contribute to the accurate prediction of driver fatigue. ResNet18’s ability to learn and distill these features into a compact vector of 256 elements is crucial for the subsequent stages of the model, where these features are combined with those extracted from physiological signals to make a final prediction about the driver’s state of alertness.

Feature combination and coupling

To achieve multimodal prediction for driver fatigue detection, this study employs distinct methods to extract features from different types of data: a time series encoder for the EEG, ECG, EMG, and EOG signals, and an image encoder for facial data. These extracted features are then used to implement multimodal predictions.

Feature combination approach As shown in Fig. 7, the first approach involves combining the features extracted from the EEG, ECG, EMG, and EOG signals with those extracted from facial data. This combined feature set is then fed into a multi-layer fully connected neural network for prediction. This method effectively harnesses the strengths of different modalities, leveraging the combined features to predict driver fatigue. By integrating physiological signals from the time series encoder with facial features from the image encoder, the model can provide a comprehensive understanding of the driver’s fatigue state.

Feature coupling approach In addition to the feature combination approach, an alternative method of feature coupling is proposed. In this approach, the time series encoder extracts n features as a time series analysis feature vector. Concurrently, the image encoder extracts an equivalent number of facial features, resulting in a facial feature vector. These two vectors are then coupled through a dot product operation, producing a coupled vector of length n . This coupled vector is subsequently input into a fully connected neural network for prediction.

As shown in Fig. 8, the feature coupling approach offers an advantage over the simple combination of features by establishing a deeper connection between physiological signals and facial expressions. By coupling features such as EEG, ECG, EMG, and EOG signals with facial data, the model creates a more integrated representation of the driver’s fatigue state. This method enhances the interaction between different modalities, potentially leading to more accurate and nuanced predictions of driver fatigue.

Multimodal feature combination neural network

The first model implemented in this study is the multimodal feature combination model, which integrates features from both image and time-series data. This model employs a ResNet18 architecture as the image encoder and a Long Short-Term Memory (LSTM) network as the time-series encoder, creating a synergistic approach for driver fatigue prediction.

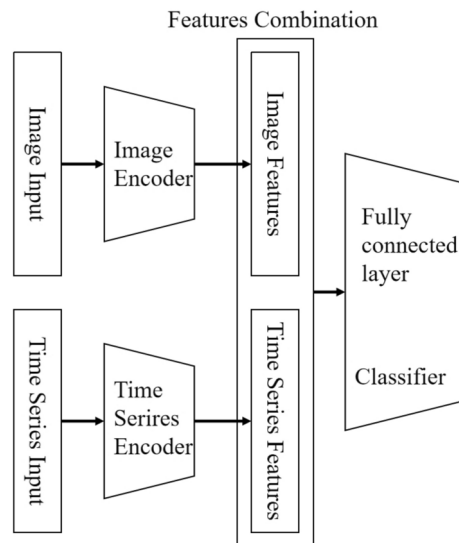


Fig. 7. Features combination.

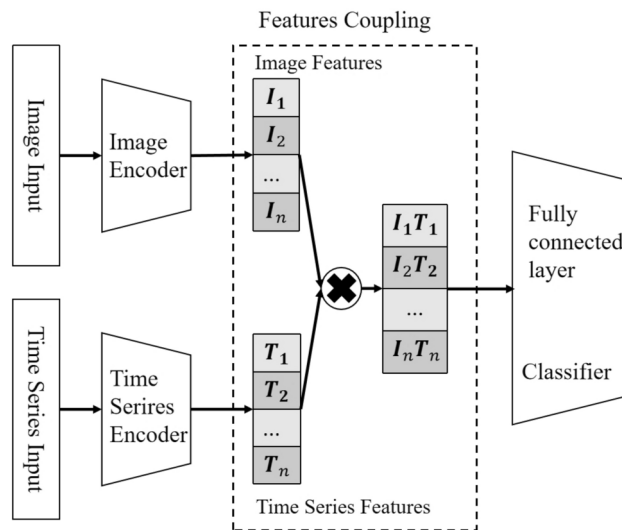


Fig. 8. Features coupling.

Typically, ResNet18 is configured to accept an input image of size (3 channels $\times$ 224 pixels $\times$ 224 pixels). However, for the specific needs of this study, the input size is adjusted to (1 channel $\times$ 75 pixels $\times$ 75 pixels). Additionally, a significant modification is applied to the feature output layer of ResNet18, setting the neuron count in the final layer to 256. This alteration ensures that the output feature vector from ResNet18 is precisely structured for integration with time-series features, as depicted in Fig. 9.

The LSTM network is utilized to process time-series data comprising 512 timesteps and 9 channels. A single-layer LSTM with 128 cells is employed, which outputs 256 time-series features. This configuration of the LSTM is designed to effectively capture the temporal dynamics within the physiological signals, including EEG, ECG, EMG, and EOG.

Upon extraction, the 256 features from ResNet18 and the 256 time-series features from the LSTM are each subjected to min-max normalization, which scales the feature values to the range of [0,1]. After normalization, these features are concatenated, resulting in a combined feature vector of length 512. This combined vector is then fed into a fully connected linear layer containing 512 neurons. The output of this layer consists of three values corresponding to different fatigue states. The classification layer is designed to categorize the driver's fatigue state into one of three levels: Alert, Low Vigilant, or Drowsy, based on the combined features from both image and time-series data. In summary, the architecture of the multimodal feature combination neural network is shown in Fig. 10.

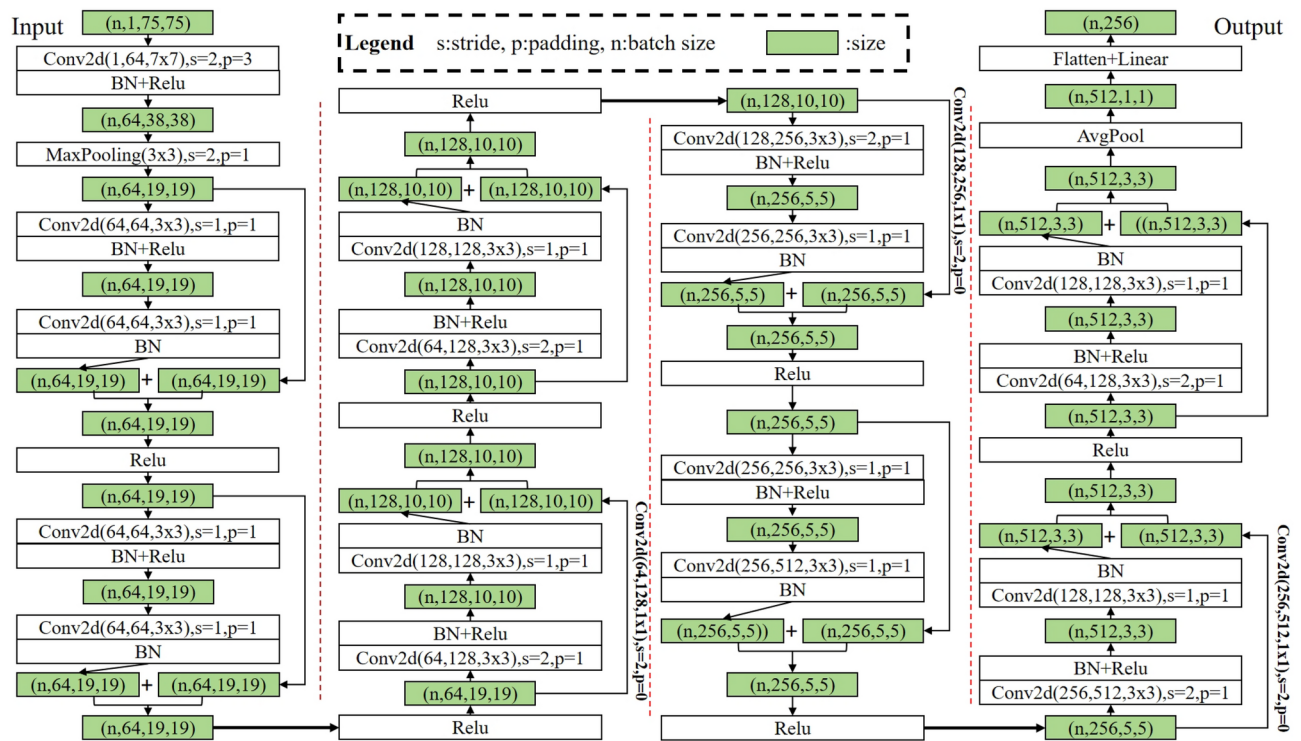


Fig. 9. Adjusted ResNet18.

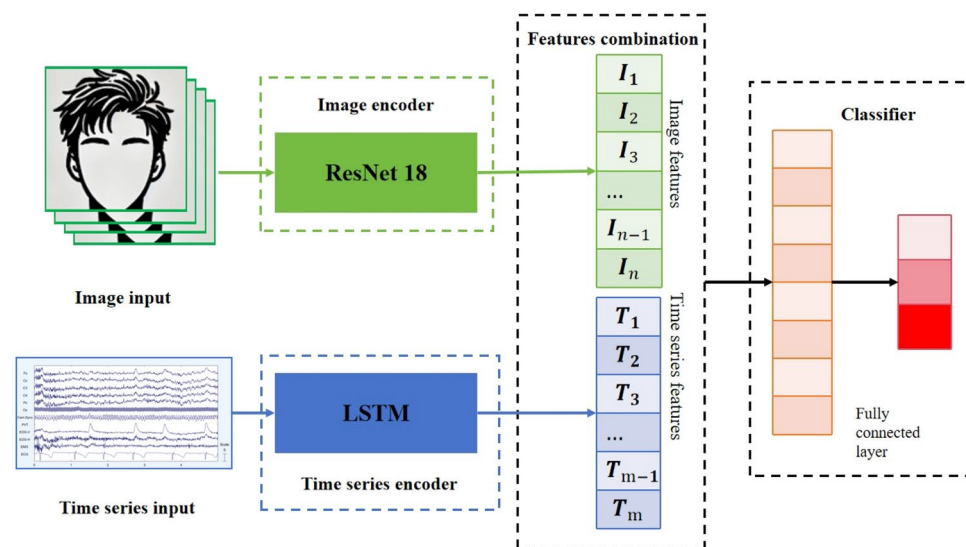


Fig. 10. Architecture of the multimodal feature combination neural network.

Multimodal feature coupled neural network

Following the development of the multimodal feature combination model, a second model is introduced, termed the multimodal feature coupled model. Unlike the first model, which merely concatenates features from multiple modalities, this model establishes a deeper linkage by multiplying the features of two different modalities. It effectively couples physiological signal features with facial signal features, thereby creating an integrated connection between EEG, ECG, EMG, EOG signals, and facial expressions.

In the feature coupled model, the same image encoder and time-series encoder as in the multimodal feature combination model are utilized, namely the adjusted ResNet18 and LSTM networks. However, a key distinction in this model is the configuration of both the LSTM and ResNet18 to output 512 features each. Before applying the dot product operation, each of these feature vectors is subjected to min–max normalization, which scales the feature values to the range of [0,1]. Subsequently, the dot product operation is applied to these two

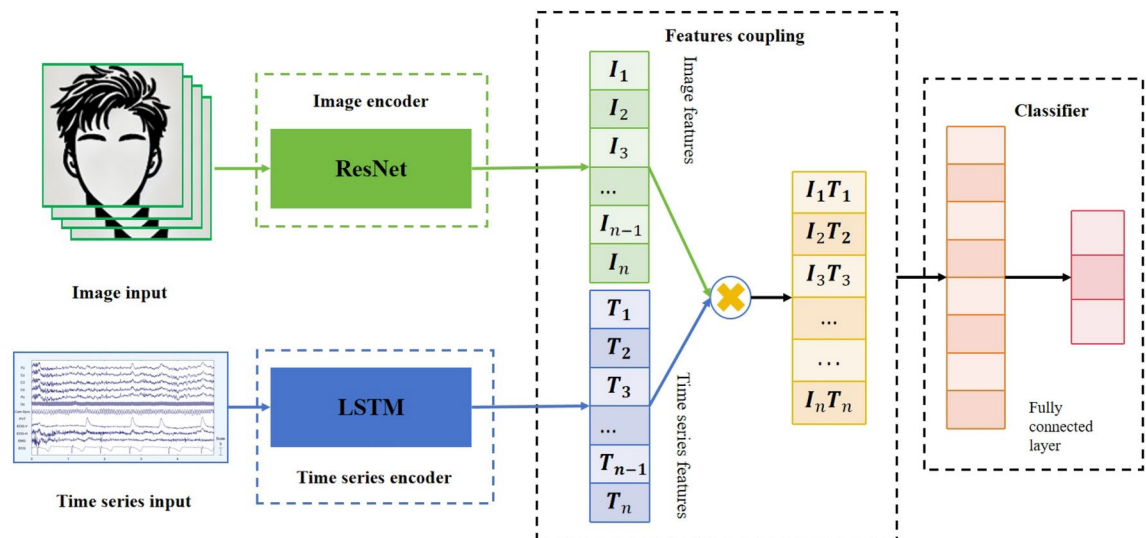


Fig. 11. Architecture of the multimodal feature coupled neural network.

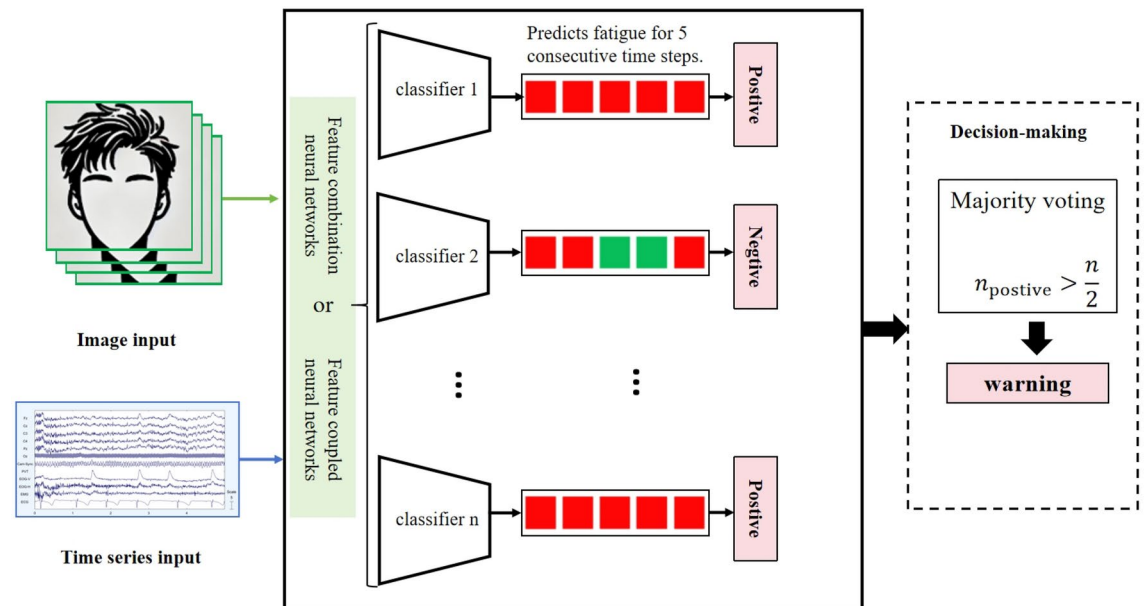


Fig. 12. Decision-making process.

normalized feature vectors, yielding a coupled feature vector of length 512. This process integrates the temporal information captured by the LSTM with the spatial information obtained from ResNet18, thereby enhancing the representation of the data.

Once the coupled feature vector is obtained, it is fed into a fully connected linear layer comprising 512 neurons. The final output layer of this model classifies the driver's fatigue state into one of three categories: Alert, Low Vigilant, or Drowsy. This classification is based on the comprehensive information encapsulated in the coupled features, providing a nuanced understanding of the driver's condition.

Figure 11 illustrates the complete architecture of the multimodal feature coupled model. This representation underscores the integration of the two modalities through feature multiplication, showcasing the model's capacity to leverage the interconnectedness of physiological and facial data for enhanced fatigue state prediction.

Decision-making mechanism

Figure 12 illustrates the design and practical application of the feature combination and feature coupled models for driver fatigue monitoring. These models aim to detect and monitor driver fatigue continuously using video and PSG (polysomnography) signals. In practice, the system samples facial and physiological signals at a

frequency of 0.1 s, with each sample covering a 1-s data window. This data window is continuously input into the model to predict the driver’s fatigue level.

To enhance the practicality and reliability of the driver fatigue monitoring system, we introduced a majority voting strategy and employed multiple feature combination or feature coupled models as classifiers. The advantage of the majority voting strategy lies in its ability to significantly improve the accuracy and robustness of the system by integrating the predictions of multiple classifiers. Each classifier evaluates its prediction over multiple consecutive time steps to ensure stability and accuracy. Specifically, for each 1-s data window, the classifier makes a prediction every 0.1 seconds. If fatigue signs are detected in all 5 consecutive predictions (i.e., within 0.5 s), the classifier outputs a “positive” state; otherwise, it outputs a “negative” state. This mechanism effectively avoids false alarms caused by transient fluctuations or noise interference, responding only to sustained fatigue signals.

The decision rule for the majority voting strategy is as follows: suppose there are n classifiers, each predicting the fatigue state of a sample as $y_i \in \{0, 1\}$, where 1 indicates “positive” (fatigue detected) and 0 indicates “negative” (no fatigue detected). The final prediction result y_{final} is calculated using the following formula:

$$y_{\text{final}} = \begin{cases} 1 & \text{if } \sum_{i=1}^n y_i \geq \frac{n}{2}, \\ 0 & \text{if } \sum_{i=1}^n y_i < \frac{n}{2}, \end{cases}$$

where $\sum_{i=1}^n y_i$ is the sum of the predictions from all classifiers, and $\frac{n}{2}$ is the half-threshold of the total number of classifiers. If more than half of the classifiers (i.e., at least $\frac{n}{2}$) predict a “positive” state, the system outputs a “positive” state; otherwise, it outputs a “negative” state. This strategy reduces the risk of misclassification by integrating the results from multiple classifiers, thereby improving the reliability and robustness of the system.

When $y_{\text{final}} = 1$, the fatigue detection system issues a warning. The majority voting mechanism effectively reduces the risk of false alarms (incorrect fatigue warnings) and missed detections (failure to detect fatigue), ensuring more accurate decisions.

This system ensures timely and reliable alerts to the driver, encouraging preventive actions such as taking breaks or finding a safe place to park when fatigue is detected. The cumulative nature of the decision-making process enhances the robustness of the system, making fatigue detection both accurate and practical in real-world driving conditions. Moreover, the continuous integration of video and physiological data ensures the effectiveness of the feature combination and feature coupled models in real-time driver monitoring, helping to reduce traffic accidents and enhance road safety.

Experimental environment and configuration

The experimental platform and settings for this study were meticulously configured to ensure optimal performance. The software environment included a 64-bit Windows 10 operating system, chosen for its compatibility and stability. Python 3.9 was utilized as the programming language due to its robust support for data processing and machine learning tasks. The integrated development environment (IDE) used was PyCharm 2023.1, selected for its efficiency in code writing and debugging. For deep learning tasks, PyTorch 1.12 was employed, recognized for its powerful capabilities and strong community support. Image processing was managed using OpenCV 4.5.5, while dlib 19.22 was used for facial detection and feature extraction. CUDA 11.3 was leveraged to accelerate the training of deep learning models by utilizing GPU computational power.

The hardware configuration featured a system powered by an AMD Ryzen Threadripper 3990X 64-Core Processor, designed to handle complex computational tasks. The system was equipped with 256 GB of RAM, ensuring smooth performance during large-scale data processing and multitasking. The NVIDIA GeForce RTX 3060 graphics card, with 12 GB of VRAM, was selected for its ample CUDA cores, supporting accelerated deep learning computations. Additionally, a 1 TB NVMe solid-state drive (SSD) was used to provide rapid data read and write speeds.

Hyperparameter tuning

Hyperparameter tuning in this study was conducted through a recursive training and validating process to optimize performance and prevent overfitting. The best hyperparameters selected for the Multimodal Feature Combination and Coupled Models are summarized in Table 1.

Hyperparameter	Combination model	Coupled model
Learning rate	0.001	0.001
Optimizer	Adam	Adam
Batch size	32	32
Number of epochs	100	100
Dropout rate	0.5	0.4
LSTM specific parameters	1 Layer, 128 Cells	1 Layer, 128 Cells
ResNet18 specific parameters	Pre-trained, Output: 256	Pre-trained, Output: 512
Loss function	Cross-Entropy Loss	Cross-Entropy Loss
Regularization	L2 Regularization: 0.0001	L2 Regularization: 0.0001
Early stopping	Patience: 100 epochs	Patience: 100 epochs

Table 1. Hyperparameters for the multimodal feature combination and coupled models.

Overall, these settings are designed to ensure the efficiency and stability of neural network training. By adopting appropriate parameter settings and leveraging the capabilities of PyTorch and CUDA, the training process is accelerated, leading to enhanced performance of the model in accurately predicting driver fatigue.

Results and discussion
Experimental results of classification models

According to “Dataset splitting” section, this study employed a subject-based data splitting method to ensure the reliability of model evaluation. For each training, validation, and testing process, data from 14 subjects were used, with 13 subjects randomly selected for training and validation. Specifically, the training set consisted of data from 9 subjects, the validation set from 4 subjects, and the independent test set from the remaining 1 subject. This method effectively prevents data leakage and overfitting, thereby ensuring the model’s generalization capability.

We then trained two models: the Multimodal Feature Combination Model and the Multimodal Feature-Coupled Model, ensuring that both met the requirements outlined in previous sections. These models were trained using the aforementioned subject-based data splitting method, focusing on optimizing each model’s parameters and performance through rigorous training on the training set and validation on the validation set.

To further enhance the robustness of the evaluation, the entire process was repeated 10 times, with a different combination of subjects selected in each run. The final performance was calculated as the average of the results from these 10 independent tests.

During the training phase, each model was trained for up to 10,000 iterations to thoroughly explore the parameter space. Additionally, an early stopping mechanism was implemented to prevent overfitting. Four key evaluation metrics-Precision, Recall, F1-score, and Accuracy-were used to assess model performance. Throughout the training process, the models achieved over 99.9% for all four evaluation metrics on the training set, demonstrating exceptional performance on the training data.

After training, the models were further evaluated on the validation and test sets. The validation set was used for hyperparameter tuning to ensure the models could effectively adapt to unseen data, preventing overfitting. The test set served as the final independent evaluation set, with the data from the remaining subject completely excluded from the training and validation processes. The key advantage of using this independent test set is that it provides an objective evaluation of model performance, as the data has not been involved in any way in the training or validation process, thus preventing data leakage. The independent test set truly reflects the models’ generalization ability on completely new data.

In summary, the use of an independent test set, along with 10 independent training and testing runs, ensured fairness and reliability in the model evaluation. This methodology confirms that the models can generalize well across different subject data and perform robustly in real-world applications.

Table 2 presents the confusion matrix used to evaluate the classification performance of the proposed model. The matrix summarizes the prediction results across three classes, where T_{ij} represents the number of instances where the actual class i was correctly or incorrectly classified as class j . Specifically:

- T_{11} , T_{22} , and T_{33} denote the number of correct predictions for Classes 1, 2, and 3, respectively.
- F_{ij} ($i \neq j$) indicates the number of misclassifications, where instances of actual class i are incorrectly predicted as class j .

Accuracy, as a global metric, measures the overall correctness of the model across all classes. It is defined as the ratio of correctly predicted instances to the total number of instances in the dataset. For a three-class problem, Accuracy is calculated as:

$$Accuracy = \frac{T_{11} + T_{22} + T_{33}}{n_{total}}, \tag{11}$$

$$n_{total} = T_{11} + T_{22} + T_{33} + F_{12} + F_{13} + F_{21} + F_{23} + F_{31} + F_{32}. \tag{12}$$

However, Precision and Recall need to be calculated for each class individually. Precision measures the accuracy of positive predictions for a specific class, while Recall assesses how well the model identifies actual instances of that class. For each class, these are defined as:

$$P_1 = \frac{T_{11}}{T_{11} + F_{21} + F_{31}}, \tag{13}$$

$$P_2 = \frac{T_{22}}{T_{22} + F_{12} + F_{32}}, \tag{14}$$

Actual	Prediction		
	Predict class 1	Predict class 2	Predict class 3
Actual class 1	T_{11}	F_{12}	F_{13}
Actual class 2	F_{21}	T_{22}	F_{23}
Actual class 3	F_{31}	F_{32}	T_{33}

Table 2. Confusion matrices.

$$P_3 = \frac{T_{33}}{T_{33} + F_{13} + F_{23}}, \tag{15}$$

$$R_1 = \frac{T_{11}}{T_{11} + F_{12} + F_{13}}, \tag{16}$$

$$R_2 = \frac{T_{22}}{T_{22} + F_{21} + F_{23}}, \tag{17}$$

$$R_3 = \frac{T_{33}}{T_{33} + F_{31} + F_{32}}. \tag{18}$$

To calculate the overall Precision and Recall, the weighted coverage method is applied. This method uses the proportion of the actual number of instances in the three categories as weights:

$$Precision = \frac{T_{11} + F_{12} + F_{13}}{n_{total}} P_1 + \frac{T_{22} + F_{21} + F_{23}}{n_{total}} P_2 + \frac{T_{33} + F_{31} + F_{32}}{n_{total}} P_3, \tag{19}$$

$$Recall = \frac{T_{11} + F_{12} + F_{13}}{n_{total}} R_1 + \frac{T_{22} + F_{21} + F_{23}}{n_{total}} R_2 + \frac{T_{33} + F_{31} + F_{32}}{n_{total}} R_3. \tag{20}$$

The F1-score, which provides a balance between Precision and Recall, is calculated as:

$$F1 - Score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}. \tag{21}$$

Table 3 presents the detailed performance report for the multimodal feature combination model across 10 independent training iterations. Both the validation set and the test set were evaluated independently to ensure a robust and comprehensive assessment of the model’s generalization capabilities. As outlined in the previous sections, both the validation set and the test set were evaluated independently to ensure a thorough and comprehensive assessment of the model’s generalization capabilities.

The results indicate a strong and consistent performance across both datasets. On the validation set, the model achieves an average accuracy of 95.10%, with a precision of 95.43%, recall of 95.19%, and an F1-score of 95.19%. These metrics reflect the model’s ability to effectively classify fatigue states with high reliability across different validation runs. Similarly, the test set performance is highly consistent with the validation set. The average accuracy on the test set is 94.87%, with a precision of 95.18%, recall of 95.04%, and an F1-score of 95.00%. The slight numerical differences between the validation and test set metrics are minimal, suggesting that the model can generalize well to unseen data while maintaining high classification performance. This is a critical indicator of the model’s robustness and suitability for real-world applications, where unseen data is common.

The overall consistency between the validation and test sets across all key performance indicators (accuracy, precision, recall, and F1-score) underscores the model’s reliability. With both sets yielding near-identical metrics, the multimodal feature combination model demonstrates strong generalization capabilities, making it a valuable tool in detecting driver fatigue in various operational environments.

Moreover, the model’s stability across 10 independent training runs further supports its robustness. The standard deviation in performance across runs is minimal, which implies that the model’s performance is not significantly affected by the random initialization or the particular training instances used in each iteration. This ensures that the model delivers consistent and dependable results even when retrained in different scenarios.

Table 4 presents the performance report for the multimodal feature-coupled model across 10 iterations of cross-validation. This model demonstrates exceptional classification performance, consistently maintaining high metrics on both the validation set and the test set.

Run	Validation set				Test set			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
1	0.9636	0.9675	0.9664	0.9650	0.9498	0.9503	0.9482	0.9511
2	0.9557	0.9570	0.9496	0.9502	0.9497	0.9531	0.9491	0.9513
3	0.9458	0.9546	0.9486	0.9506	0.9502	0.9276	0.9270	0.9279
4	0.9568	0.9622	0.9572	0.9604	0.9649	0.9632	0.9632	0.9587
5	0.9503	0.9589	0.9550	0.9565	0.9443	0.9501	0.9446	0.9446
6	0.9638	0.9610	0.9621	0.9601	0.9458	0.9474	0.9474	0.9467
7	0.9492	0.9523	0.9502	0.9511	0.9552	0.9571	0.9571	0.9579
8	0.9303	0.9303	0.9296	0.9289	0.9517	0.9501	0.9501	0.9501
9	0.9497	0.9553	0.9544	0.9509	0.9668	0.9692	0.9692	0.9672
10	0.9446	0.9450	0.9449	0.9457	0.9461	0.9450	0.9450	0.9445
Average	0.9510	0.9543	0.9519	0.9519	0.9487	0.9518	0.9504	0.9500

Table 3. Performance report of the multimodal feature combination model.

Run	Validation set				Test set			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
1	0.9831	0.9832	0.9831	0.9832	0.9836	0.9835	0.9832	0.9838
2	0.9848	0.9849	0.9848	0.9849	0.9748	0.9747	0.9748	0.9747
3	0.9863	0.9844	0.9863	0.9864	0.9759	0.9758	0.9759	0.9758
4	0.9830	0.9833	0.9830	0.9827	0.9923	0.9902	0.9903	0.9902
5	0.9880	0.9881	0.9880	0.9881	0.9746	0.9745	0.9746	0.9746
6	0.9817	0.9818	0.9817	0.9818	0.9836	0.9835	0.9835	0.9835
7	0.9862	0.9863	0.9862	0.9863	0.9892	0.9891	0.9892	0.9891
8	0.9872	0.9873	0.9872	0.9873	0.9897	0.9896	0.9897	0.9896
9	0.9902	0.9903	0.9902	0.9902	0.9905	0.9904	0.9905	0.9904
10	0.9919	0.9920	0.9919	0.9919	0.9867	0.9866	0.9867	0.9866
Average	0.9862	0.9862	0.9862	0.9863	0.9841	0.9838	0.9839	0.9838

Table 4. Performance report of the multimodal feature-coupled model.

The validation set results are particularly impressive, with the model achieving an average accuracy of 98.62%, precision of 98.62%, recall of 98.62%, and an F1-score of 98.63%. These metrics show very little variation across the different runs, indicating the robustness and reliability of the multimodal feature-coupled model in detecting fatigue states. The high accuracy across all metrics suggests that the model effectively leverages the integration of physiological signals and facial expressions, resulting in a highly capable fatigue detection system.

Similarly, the test set results also demonstrate strong performance. The average accuracy for the test set is 98.41%, with a precision of 98.38%, recall of 98.39%, and an F1-score of 98.38%. The minimal difference between the validation and test set results underscores the model's strong generalization ability. This consistency between the two datasets highlights that the model is not only effective on data it has seen during training but also performs well on entirely unseen data, which is critical for real-world applications.

The overall performance of the multimodal feature-coupled model is highly commendable. The close alignment of the metrics across both the validation and test sets confirms the model's generalization strength and its reliability in accurately detecting driver fatigue. Unlike models that may overfit the training data, this feature-coupled model maintains high accuracy, precision, recall, and F1-scores across different datasets.

In conclusion, the multimodal feature-coupled model demonstrates advanced capabilities in predicting driver fatigue. By integrating physiological signals and facial expressions, it achieves high classification accuracy, making it a promising tool for enhancing road safety through real-time fatigue detection. The model's consistent performance across different iterations suggests its potential for practical applications in systems requiring reliable and accurate detection of fatigue.

Ablation studies

To validate the contributions of different components in the proposed multimodal driver fatigue detection model, a series of ablation studies were conducted. The objective of these experiments was to assess the impact of removing or altering key components—namely, the LSTM network, ResNet18, and the feature coupling mechanism—on the model's overall performance. For each ablation experiment, 10 independent training and validation runs were conducted, and the average performance metrics were calculated to ensure robust and reliable results.

The baseline model, which serves as the benchmark for comparison, utilizes an LSTM network to process physiological signals, ResNet18 for image feature extraction, and a feature coupling mechanism to integrate the multimodal features for fatigue prediction. This complete model represents the full capabilities of the proposed approach.

In the first ablation model, the LSTM network was removed, and only the ResNet18-extracted image features were used for prediction. This experiment aimed to evaluate the contribution of temporal physiological signals to the model's overall performance. Meanwhile, the second ablation model excluded the ResNet18 component, relying solely on the LSTM-extracted physiological signal features for prediction. The purpose of this experiment was to assess the importance of facial image data in detecting fatigue. Furthermore, in the third ablation model, both the LSTM and ResNet18 components were retained, but the feature coupling mechanism was removed. Instead, the features were simply concatenated before prediction. This experiment evaluated the effect of the coupling mechanism on the integration of multimodal features.

Table 5 presents the performance of each model in terms of accuracy, precision, recall, and F1-score.

The results from the ablation studies offer profound insights into the underlying mechanisms that drive the effectiveness of the proposed multimodal driver fatigue detection model. By systematically removing or altering key components—LSTM, ResNet18, and the feature coupling mechanism—we can better understand how each contributes to the model's ability to accurately predict driver fatigue.

Starting with the baseline model, which leverages a sophisticated feature coupling mechanism, the model achieves near-perfect performance across all metrics, with an accuracy of 98.41%, a precision of 98.38%, a recall of 98.39%, and an F1-score of 98.38%. This exceptional performance underscores the effectiveness of the feature coupling approach in integrating multimodal data. Specifically, the coupling mechanism allows the model to capture and exploit the intricate relationships between facial expressions and physiological signals, resulting in

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Baseline model (feature coupled)	98.41	98.38	98.39	98.38
Ablation model 1 (no LSTM)	88.51	88.66	88.83	88.42
Ablation model 2 (no ResNet18)	90.11	91.78	89.75	89.66
Ablation model 3 (feature combination)	94.87	95.18	95.04	95.00

Table 5. Performance of ablation models.

a highly accurate and robust prediction system. Moreover, the high F1-score further indicates that the model maintains a remarkable balance between precision and recall, minimizing both false positives and false negatives.

When examining Ablation Model 1, where the LSTM network was removed, a significant decline in performance is observed, with accuracy dropping to 88.51% and corresponding decreases in precision, recall, and F1-score. This drastic reduction highlights the critical role of the LSTM network in processing temporal physiological data, such as EEG, ECG, and EMG signals. The LSTM’s ability to model long-term dependencies and temporal sequences is essential for capturing the progressive nature of driver fatigue, which often manifests subtly over time. Without LSTM, the model loses its ability to interpret these temporal dynamics effectively, resulting in less accurate fatigue detection. This suggests that physiological data alone, when not temporally processed, lacks the context necessary for accurate fatigue assessment, leading to a substantial performance degradation.

Interestingly, Ablation Model 2, where ResNet18 is removed, shows that the model’s precision slightly improves to 91.78%, but this comes at the cost of lower accuracy (90.11%), recall (89.75%), and F1-score (89.66%). This outcome reveals the nuanced role of facial image data in fatigue detection. While the model can still perform reasonably well without ResNet18, indicating that physiological signals provide significant predictive power, the visual cues extracted by ResNet18 are crucial for maintaining comprehensive and consistent performance. The improvement in precision suggests that, without facial features, the model may become more conservative in predicting fatigue, leading to fewer false positives but potentially missing more true positives (as indicated by the lower recall). This trade-off between precision and recall points to the complementary nature of visual and physiological data—each modality enhances the model’s ability to generalize across different scenarios, and their combined use through ResNet18 ensures a more balanced and reliable detection system.

In Ablation Model 3, the feature coupling mechanism was replaced with a simpler feature combination approach, resulting in a performance drop, with a accuracy at 94.87%, a precision of 95.18%, a recall of 95.04%, and an F1-score of 95.00%. Although these metrics remain high, the decrease compared to the baseline model demonstrates the superior effectiveness of the feature coupling mechanism over simple feature concatenation. The coupling mechanism not only integrates the multimodal features but also enhances their interaction, allowing the model to capture complex, non-linear relationships between the facial and physiological data. This richer interaction leads to better generalization and a higher overall predictive accuracy. The drop in all performance metrics in the absence of the coupling mechanism suggests that this component is crucial for fully leveraging the combined information from different modalities, confirming that sophisticated feature integration is a key driver of the model’s success.

In summary, the ablation studies reveal that each component—LSTM, ResNet18, and the feature coupling mechanism—contributes uniquely to the model’s performance. The LSTM network’s role in processing temporal data is irreplaceable, as it captures the evolution of physiological signals over time, which is vital for detecting the gradual onset of fatigue. The ResNet18 network’s ability to extract detailed facial features ensures that the model can account for immediate visual cues that may indicate fatigue, such as eye closure and facial expressions. Finally, the feature coupling mechanism enhances the model’s ability to integrate and interpret multimodal data, creating a synergistic effect that leads to superior accuracy and reliability.

These findings validate the design of the proposed multimodal model, where the combination of LSTM, ResNet18, and feature coupling is essential for capturing the full spectrum of indicators associated with driver fatigue. The model’s ability to perform well across different metrics, even when certain components are removed, highlights its robustness, but the superior performance of the complete model underscores the importance of an integrated approach that leverages the strengths of each component. This integrated approach not only improves the accuracy of fatigue detection but also enhances the model’s generalization capability, making it more reliable for real-world applications.

Model comparison analysis

Table 6 delineates the results of an extensive evaluation, encompassing twenty models from a range of studies leveraging the DROZY dataset, alongside two innovative models conceived in the present research, denoted as Models 21 and 22. The pivotal metric for this comparative analysis is the predictive accuracy of the models regarding the detection of fatigue states.

The models listed in Table 6 can be bifurcated into two categories: multimodal and single-modal approaches. Specifically, Models 14, 15, and 20, originating from distinct research endeavors, adopt a multimodal strategy, integrating various data types to enhance predictive performance. In contrast, Models 21 and 22, which are central to this study, advance the field with the introduction of a multimodal feature combination neural network and a multimodal feature coupled neural network, respectively. These models aim to synergize data from multiple sources to improve the accuracy of fatigue state predictions. All other models are single-modal models, relying solely on data from a single source within the DROZY dataset for model development and

Index	Method and model	Raw data type	Accuracy (%)
1	VGG-19 ⁴⁶	Facial image	85.00
2	ResNet ⁴⁶	Facial image	79.75
3	VGC-Face ⁴⁶	Facial image	88.97
4	Electrocardiographic RR interval statistical features + RBF ⁴⁷	Time series (ECG)	81.96
5	MobileNets ⁴⁸	Facial image	85.68
6	Frame average + MobileNets ⁴⁸	Video	91.07
7	EEG spectrogram + Capsule network ¹⁵	Time series (ECG)	86.74
8	Handcrafted features + XGBoost classifier ³⁵	Time series (ECG)	82.10
9	Handcrafted (advanced) features + XGBoost classifier ³⁵	Time series (ECG)	93.40
10	CNN-BiLSTM feature extraction + XGBoost classifier ³⁵	Time series (ECG)	91.60
11	Handcrafted (advanced) features + CNN-BiLSTM feature extraction + XGBoost classifier ³⁵	Time series (ECG)	95.60
12	Hybrid attention cascade network + XGBoost classifier ³⁵	Video	97.90
13	Deep residual networks ¹⁴	Time series (EEG)	91.00
14	Hybrid of ECG and video-based detection method ³⁵	Time series (EEG) + Video	98.10
15	Multimodal approach of video and ECG signals ³⁵	Time series (EEG) + Video	99.60
16	Handcrafted features + SVM ⁴⁹	Video	94.40
17	Handcrafted features + Physiological model (KNN) ³⁸	Time series (EEG, EOG, EMG)	91.43
18	Handcrafted features + Physiological model (MLP) ³⁸	Time series (EEG, EOG, EMG)	76.70
19	Behavioral model (ResNet101 + LSTM) ⁶	Facial image	77.42
20	Hybrid Model: Physiological model (KNN) + behavioral approaches (ResNet101 + LSTM) ³⁸	Time series (EEG, EOG, EMG)+Facial image	93.10
21	Multimodal feature combination neural network	Time series (EEG, EMG, ECG, EOG) + Facial image	94.87
22	Multimodal feature coupled neural network	Time series (EEG, EMG, ECG, EOG) + Facial image	98.41

Table 6. Comparison among drive fatigue detection models using the DROZY database.

training. This includes data types such as ECG readings, facial images, and video recordings. Each model’s methodology and data utilization are critical factors that influence its effectiveness in accurately classifying fatigue states, underscoring the importance of the selected feature processing method in the development of robust predictive models.

Moreover, Table 7 includes models that integrate handcrafted features and those that leverage deep learning technologies for automatic feature extraction from raw data. Table 7 provides a detailed overview of the handcrafted features utilized by certain models. An evaluation of the results in Table 6 reveals that models trained directly on raw data, such as Models 1, 2, 3, 5, 6, 7, 13, and 19, exhibit commendable predictive accuracy, suggesting that raw data possesses inherent qualities conducive to the detection of fatigue states. Nonetheless, there exists latitude for the augmentation of model performance.

Feature selection is a crucial step in predictive models as it significantly affects the performance of the model. Interestingly, the incorporation of conventional handcrafted features—such as mean and standard deviation—into certain models does not necessarily result in a significant upturn in accuracy, as illustrated by the disparate outcomes of Models 4, 8, 16, 17, and 18. This inconsistency in performance accentuates the potential limitations of manual feature engineering, which can yield a spectrum of results ranging from high accuracy to notable underperformance.

According to Table 5, it can be seen that certain features yield better predictive performance than others. For instance, frequency domain features of ECG such as spectral powers of LF, MF, HF, and TF, as well as heart rate variability features like LF/HF, LF/TF, HF/TF, and HF/(TF-LF), are shown to be particularly effective, with models achieving high accuracy rates. Non-linear features such as sample entropy, Lyapunov Exponent (LyE), Hurst exponent (HurstE), and Detrended Fluctuation Analysis (DFA) from ECG, as well as relative medium and high-frequency components from EMG, are also indicated to be strong predictors of fatigue states, as models leveraging these features exhibit high accuracy. Models 9 and 11 involve professional optimization of single-modal models and advanced feature extraction techniques, which introduce these complex features and demonstrate the improvement in accuracy, affirming the benefits of complex data-driven feature engineering methods.

However, manual feature extraction inherently involves a level of subjectivity, as the feature selection is guided by the researcher’s intuition and understanding of the domain. This subjectivity can lead to bias, as the researcher might inadvertently select features that reflect their expectations or miss out on relevant features that an automated process could identify.

To mitigate the subjectivity and potential bias of manual feature extraction, deep learning methods offer significant advantages. Techniques such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory networks (LSTMs) are capable of automatically learning to identify and extract features from raw, high-dimensional data. CNNs are particularly adept at processing spatial information in image data, while LSTMs excel at handling temporal dependencies in time-series data, making them suitable for datasets like DROZY that contain both physiological signals and facial expressions. Deep learning techniques for automatic feature extraction, employed in Models 10, 12, and several multimodal models, have proven to be more effective, with

Model index	Features	Accuracy
4	• Mean, variance, standard deviation, geometric mean,	81.96
	harmonic mean, interquartile, kurtosis, skewness, median,	
	mode, maximal, and minimal of ECG RR interval.	
8	• Time domain:	82.10
	(1) Heart rate: features of ECG: mean, standard deviation	
	(std), skewness, kurtosis, percentage of points higher than	
	mean + std, and percentage of points lower than mean - std.	
	(2) Heart rate variability features of ECG: mean, std,	
	skewness, kurtosis, root mean square, percentage of points	
	higher than mean + std, and percentage of points lower	
	than mean - std.	
	• Frequency domain:	
	(1) Features obtained from raw ECGs: spectral power 0–6 Hz	
	(per 0.1 Hz)	
9/11	(2) Heart rate variability features of ECG: spectral powers	93.40/95.60
	of LF, MF, HF, and TF.	
	• Waveform-related features of ECG: tpr and mtptr	
	• Nonlinear Features: rcmse, Shannon entropy	
	• Time domain:	
	ECG: lag, mean period.	
16	• Frequency domain:	94.40
	Spectral power ratio of ECG: LF/HF, LF/TF, HF/TF,	
	LF/(LF-LF), and HF/(TF-LF).	
	• Nonlinear features	
17/18	ECG: sample entropy, LyE, HurstE, and DFA.	91.43/76.70
	• 15 consecutive EARs in videos	
	• EEG Pz channel: relative alpha power, sample entropy,	
	Fuzzy Entropy	
	• EOG: Eye opening/closure velocity, skewness, kurtosis,	
17/18	Delay Ratio, Delay of reopening	91.43/76.70
	• EMG: relative medium frequency, relative high	
	Frequency	

Table 7. Handcrafted features.

marked improvements in accuracy. These techniques, including the use of LSTM and CNN architectures, enable the extraction of complex, high-dimensional features that better capture the nuances of the data for fatigue state prediction.

Furthermore, the use of pre-trained models, which have already learned robust feature representations on large and diverse datasets, can accelerate the feature extraction process and enhance the performance of fatigue prediction models. These models can be fine-tuned on the specific dataset of interest, leveraging their learned knowledge to quickly and effectively capture relevant features.

The analysis suggests that while manually engineered features have shown promise, deep learning methods provide a more objective, efficient, and potentially more accurate approach to feature extraction. By leveraging LSTM, CNN, and pre-trained models, researchers can automate the feature extraction process, leading to faster development times and improved predictive performance in fatigue and drowsiness state classification.

The scholarly analysis of the data presented in Table 1 reveals a clear stratification between multimodal and single-modal predictive models in the domain of drowsiness detection using the DROZY dataset. This examination particularly focuses on models indexed as 14, 15, and 20, which represent multimodal approaches cited from various research studies, and on models 21 and 22, which are novel contributions of this research featuring a multimodal feature combination and coupling within neural network architectures, respectively.

A meticulous evaluation of feature extraction methodologies reveals that models employing a multimodal strategy exhibit superior predictive accuracy compared to their single-modal counterparts. The multimodal models not only integrate data from diverse sources but also leverage the synergistic effect of combined features, thereby enhancing the robustness and precision of the prediction.

Models 14 and 15, as well as the multimodal models 21 and 22 proposed in this study, exemplify multimodal feature fusion models that achieve extremely high accuracy. These models consistently achieve accuracy rates exceeding 95%, with average levels superior to those of the other single-modal models in Table 6. This high level of precision underscores the efficacy of multimodal models, which utilize the complementary nature of disparate data modalities to construct a more comprehensive predictive framework.

In the context of driver fatigue detection, the multimodal approach, particularly the combination of facial information with physiological signals such as EEG, EMG, EOG, and ECG, provides a comprehensive and nuanced understanding of the driver's state. This holistic approach significantly enhances the accuracy and reliability of fatigue detection systems compared to single-modality methods. Facial information, including visual cues like eye closure, yawning, and changes in head position, offers intuitive and observable indicators of fatigue. Advanced image processing techniques applied to facial data can detect subtle changes that signal the onset of drowsiness or fatigue. However, relying solely on facial information may not always be sufficient, as it can be influenced by external factors such as lighting conditions or the driver's posture.

On the other hand, physiological signals provide direct measurements of the body's internal state. EEG signals are crucial for assessing brain activity patterns related to alertness and sleep stages. EOG signals, which track eye movements, are particularly effective in detecting microsleeps, a dangerous condition where the driver momentarily loses consciousness. EMG signals reveal muscle activity, potentially indicating physical tiredness, while ECG signals can show changes in heart rate variability, often linked to fatigue.

By integrating these diverse data sources, a multimodal approach can achieve higher prediction accuracy. The combination of facial and physiological data allows for the detection of a wide range of fatigue indicators, encompassing both behavioral and biological aspects. This integration makes the system more robust to noise and variability in individual data sources, ensuring reliable detection even under challenging conditions.

It is worth noting that most multimodal models simply combine the features of different data sources before the prediction process. This is in contrast to the approach taken by Model 20, which represents a model-level fusion, wherein two distinct models, one physiological and the other behavioral, are independently trained. The predictive outputs of these models are subsequently aggregated to yield a collective result. This methodological distinction is pivotal, as it underscores the fundamental differences between combining features at the data level and amalgamating the outputs of distinct models. Such a strategy contrasts with the feature-level coupling employed by Models 21 and 22, which intertwines features at an earlier stage, allowing for a more integrated analysis and potentially capturing more complex interactions between different data modalities.

The research presented in this paper includes Model 21, which employs a modality fusion approach similar to Models 14 and 15, utilizing deep learning techniques to extract features and directly combine them into feature vectors for classification. These models demonstrate exceptional performance, particularly Model 15, with an accuracy rate exceeding 99%. The distinction lies in the feature extraction sources: Models 14 and 15 derive features from video and ECG data streams, demanding significant computational resources related to video data processing. In contrast, Model 21 extracts features from facial images, significantly reducing computational requirements. This approach not only showcases the efficiency of utilizing facial images for feature extraction but also emphasizes the potential of deep learning in enhancing model performance with reduced computational demands. Model 22, on the other hand, demonstrates the advantages of processing image data, which is less computationally intensive than video data, while still achieving a remarkable accuracy rate of 98.41%. This model exemplifies the benefits of the proposed approach, which not only ensures high predictive performance but also maintains computational feasibility.

Model 21 adopts a multimodal feature combination framework that integrates distinct features from various data modalities, such as facial images and physiological signals like EEG and ECG. This integration allows the model to perform a comprehensive analysis by considering a wide spectrum of fatigue indicators, enhancing the predictive capability of the model. By drawing on multiple data sources, the model is not solely reliant on a single type of data, making it more versatile and robust against anomalies or noise specific to any one data modality. This multimodal feature combination allows for the capture of different aspects of fatigue, such as the visible signs of drowsiness in facial expressions and the underlying physiological states reflected in EEG and ECG signals.

The strength of Model 21 lies in its ability to aggregate diverse features into a cohesive predictive framework. For instance, while facial images provide direct visual cues such as eye closure or yawning, which are immediate indicators of drowsiness, physiological signals like EEG and ECG offer insight into the driver's internal state, capturing the subtle shifts in brain activity and heart rate that precede visible signs of fatigue. This dual approach ensures that the model can detect fatigue at both the behavioral and physiological levels, increasing its sensitivity and reliability across different scenarios.

However, while Model 21 effectively combines these features, Model 22 advances the concept of multimodal integration through a more sophisticated feature coupling approach. This approach not only combines features from different modalities but couples them in a way that allows the features from one modality to act as dynamic weights for the features of another. This represents a significant leap forward in the depth of interaction between different data types, enabling the model to uncover and exploit complex relationships that are not apparent when features are merely aggregated.

The feature coupling mechanism in Model 22 fundamentally transforms how multimodal data is utilized in fatigue detection. Instead of treating features from each modality as independent inputs, the coupling mechanism dynamically adjusts the influence of one set of features based on the other. For example, subtle changes in facial expressions, such as slight eyelid drooping or micro-expressions of fatigue, can be weighted more heavily if there are concurrent fluctuations in EEG or ECG signals that indicate a drop in alertness. This dynamic weighting allows the model to prioritize the most relevant features based on the current state of the driver, leading to a more accurate and context-sensitive prediction.

This dynamic interaction between facial and physiological features reflects a more holistic understanding of the driver's condition. The model effectively integrates external visual signs of fatigue with internal physiological states, ensuring that its predictions are grounded in a comprehensive view of the data. This ability to correlate and weight features across modalities adds a layer of sophistication to the model, allowing it to capture the multifaceted nature of fatigue more effectively than traditional models that rely on static feature combinations.

One of the key advantages of Model 22's coupling mechanism is its self-regulatory feature weighting, which functions like an internal system of checks and balances. This mechanism fine-tunes the predictive model by adjusting the emphasis on different features based on their relevance to the current data context. For instance, in scenarios where the facial expressions alone might be misleading due to external factors like poor lighting or unusual head angles, the model can rely more on physiological data to make an accurate prediction. Conversely, in situations where physiological signals are noisy or ambiguous, the model can give more weight to clear visual cues. This adaptability makes the model particularly robust in real-world conditions, where data quality can vary.

Furthermore, the focus on image features in Model 22 also offers a computational advantage. Processing static images is generally less resource-intensive than processing video data, which involves handling large sequences of frames. By efficiently coupling static image data with physiological signals, Model 22 achieves high predictive accuracy without the heavy computational burden typically associated with real-time video analysis. This efficiency makes the model highly suitable for real-time applications, such as in-vehicle driver monitoring systems, where both accuracy and speed are critical.

Model 22's innovative feature coupling approach marks a significant advancement in the development of fatigue detection systems. By enabling a deeper and more dynamic interaction between different data modalities, the model not only improves accuracy but also enhances adaptability and efficiency. This approach could be particularly impactful in applications where the safety and performance of individuals are at stake, such as in transportation, where timely and accurate detection of driver fatigue can prevent accidents and save lives.

In summation, the research underscores the inherent advantages of employing multimodal models, particularly those that leverage feature coupling strategies, over single-modal approaches for fatigue detection. The models developed in this study, especially Model 22, demonstrate superior performance by advancing beyond traditional feature combination methods. The innovative methodology of feature coupling within neural networks represents a substantial leap forward, significantly improving the accuracy and reliability of fatigue state predictions. This advancement not only highlights the effectiveness of integrating multiple data sources but also showcases the capabilities of neural networks to assimilate and analyze complex, multimodal data for more precise and insightful outcomes. The findings from this study affirm the critical role of advanced neural network architectures in refining feature integration processes, ultimately leading to the development of superior predictive models in the field of fatigue detection.

Conclusion and future work

Conclusion

Driver fatigue has emerged as a critical concern in traffic safety, with drowsy driving posing significant risks to both individual drivers and public safety. Advances in sensor technology and data analytics present a burgeoning opportunity to collect detailed physiological and behavioral data, paving the way for sophisticated fatigue detection models. The integration of multimodal data sources, including physiological signals and facial expressions, offers a promising approach to comprehensively capture the complexity of fatigue.

This study leverages the potential of multimodal data through advanced neural network models, utilizing the extensive information provided by the DROZY database. This dataset, which includes a wide array of physiological signals and facial features collected under conditions of increasing sleep deprivation, supports the development of more holistic and accurate fatigue detection models. The research introduces two novel neural network models: the multimodal feature combination model and the multimodal feature coupled model. These models synergize data from multiple modalities to enhance predictive accuracy, demonstrating the superiority of multimodal approaches over single-modal techniques.

The multimodal feature combination model integrates diverse data sources such as EEG, ECG, and facial images, thereby broadening the spectrum of detectable fatigue indicators and potentially improving prediction robustness. Meanwhile, the multimodal feature coupled model, which achieves an impressive accuracy of 98.41%, precision of 98.38%, recall of 98.39%, and an F1-score of 98.38%, demonstrating its exceptional capability in accurately detecting driver fatigue. This approach adaptively weights features from different modalities, offering a nuanced perspective on fatigue states. Furthermore, the focus on image data processing reduces computational demands, making the model well-suited for real-time applications, such as in-vehicle monitoring systems. In comparison, the multimodal feature combination model also performed well, with an accuracy of 94.87%, precision of 95.18%, recall of 95.04%, and an F1-score of 95.00%. To apply the model in real-world scenarios, the decision-making phase incorporates a majority voting strategy, where multiple classifiers' predictions are aggregated to ensure reliable and consistent warnings based on sustained fatigue signals. These results emphasize the advantages of the multimodal feature coupled model in overcoming the challenges of driver fatigue detection, making it a valuable tool for improving road safety through advanced, efficient monitoring systems in vehicle environments.

In summary, this research makes a contribution to the field of fatigue detection by proposing models that excel in both accuracy and computational efficiency. The findings underscore the effectiveness of neural network architectures in leveraging multimodal data for sophisticated feature integration and coupling, which promises to enhance safety and performance in settings where fatigue detection is critical.

Future work

While the proposed multimodal neural network models have demonstrated substantial advancements in the accurate detection of driver fatigue, several avenues remain open for further exploration and development. Future research should focus on enhancing the effectiveness and applicability of these models by addressing key areas that will improve their overall performance and real-world utility.

Firstly, expanding the dataset to include a more diverse range of participants across various age groups, ethnicities, and driving environments is crucial for improving the model's generalizability and robustness across

different populations. A more comprehensive dataset will ensure that the models can perform reliably across a wide array of real-world scenarios.

Secondly, implementing these models in real-time monitoring systems within vehicles presents challenges related to processing speed, latency, and system reliability, particularly under varying environmental conditions. Addressing these technical challenges will be essential for the successful deployment of fatigue detection systems in real-world driving scenarios.

Moreover, future studies could explore advanced data fusion techniques, such as attention mechanisms or graph neural networks, to capture more intricate relationships between different data modalities. These approaches could lead to even greater accuracy and reliability in detecting driver fatigue, further enhancing the system's ability to interpret complex interactions within multimodal data.

Personalization of models is another critical area for future research. Individual drivers exhibit varying physiological and behavioral responses to fatigue, and developing adaptive models that can learn and adjust to a driver's baseline alertness and fatigue thresholds would significantly improve the sensitivity and specificity of fatigue detection systems.

Furthermore, while subjective approaches like the Stanford Sleepiness Scale (SSS) are easy to administer and have been validated in various studies, their application in real-time, online scenarios is limited due to their reliance on self-reporting. This limitation makes them less suitable for continuous monitoring in dynamic environments such as driving. Future work should focus on overcoming these limitations by further advancing objective methods that do not require user input. Objective approaches, as utilized in the proposed multimodal models, can provide continuous, real-time assessments of fatigue, thereby enhancing both the practicality and accuracy of fatigue detection systems.

Additionally, integrating fatigue detection systems with autonomous driving technologies presents promising opportunities. Such integration could facilitate seamless transitions between human and autonomous driving, particularly in situations where a driver's fatigue levels are critically high, thereby enhancing overall road safety.

Lastly, the use of multimodal data in these systems raises important ethical and privacy concerns. It is imperative to develop systems that ensure data security and protect user privacy while maintaining the effectiveness of fatigue detection. Addressing these considerations is essential for gaining user trust and ensuring the responsible deployment of these technologies.

Data availability

The data used in this study is sourced from the publicly available DROZY dataset (<http://www.drozy.ulg.ac.be/>)⁴⁵. Additional data, including code and relevant datasets used in this research, are available from the corresponding author upon reasonable request.

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Author contributions

Shengli Cao: Conceptualization, Methodology, Software, Data curation, Writing—Original draft preparation. Peihua Feng: Supervision, Writing—Review and Editing, Funding acquisition. Wei Kang: Validation, Formal analysis, Investigation. Bo Wang: Visualization, Investigation, Resources.

Declarations

Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Ethical and informed consent for data used

This study was conducted in compliance with the ethical standards of the institutional research committee and in accordance with the 1964 Helsinki Declaration and its subsequent amendments or comparable ethical standards. No animals were involved in this research, and all procedures involving human participants were performed following established ethical guidelines. The DROZY dataset used in this research was collected by the Laboratory for Signal and Image Exploitation (INTELSIG) at the Department of Electrical Engineering and Computer Science, University of Liège (ULg), Belgium. According to the DROZY license agreement, the dataset was collected under appropriate ethical oversight, ensuring the anonymity and informed consent of all participants. The license agreement for the DROZY dataset can be accessed at <https://orbi.uliege.be/handle/2268/191620>. Informed consent was obtained from all individual participants included in the DROZY dataset during its initial collection. This study adheres to the terms and conditions of the DROZY license agreement, including ensuring participant anonymity and appropriately citing the original source.

Additional information

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